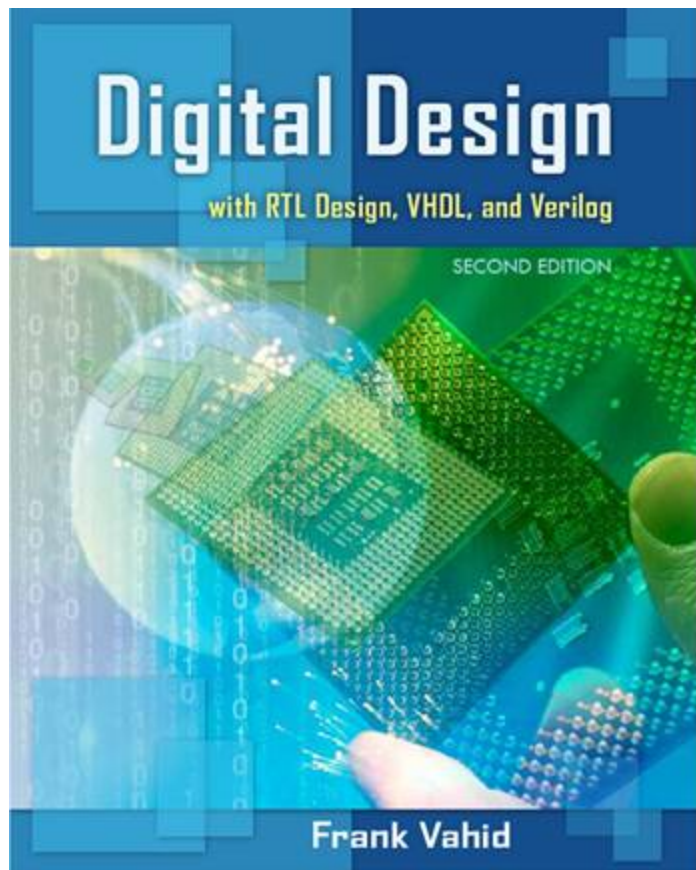




Digital Design

Chapter 5: Register-Transfer Level (RTL) Design

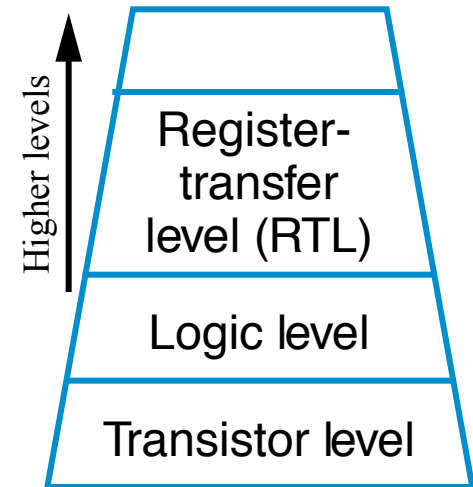
Slides to accompany the textbook *Digital Design, with RTL Design, VHDL, and Verilog*, 2nd Edition,
by Frank Vahid, John Wiley and Sons Publishers, 2010.
<http://www.ddvahid.com>

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Introduction

- Chpt 2
 - Capture Comb. behavior: Equations, truth tables
 - Convert to circuit: AND + OR + NOT → Comb. logic
- Chpt 3
 - Capture sequential behavior: FSMs
 - Convert to circuit: Register + Comb. logic → Controller
- Chpt 4
 - Datapath components, simple datapaths
- Chpt 5
 - Capture behavior: High-level state machine
 - Convert to circuit: Controller + Datapath → Processor
 - Known as “RTL” (register-transfer level) design



Levels of digital design abstraction

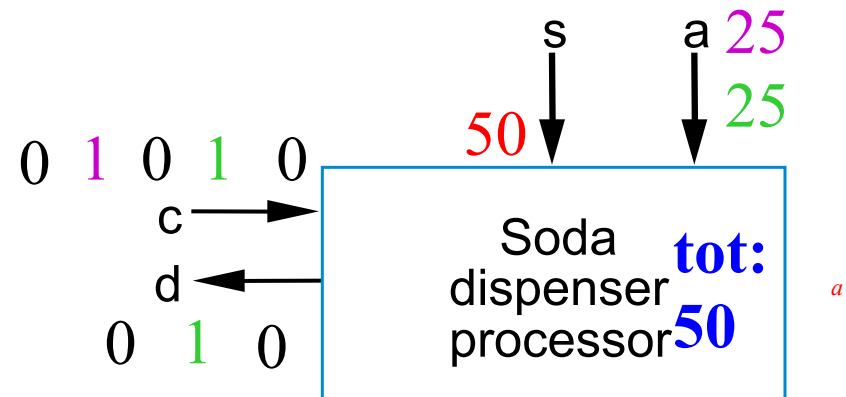
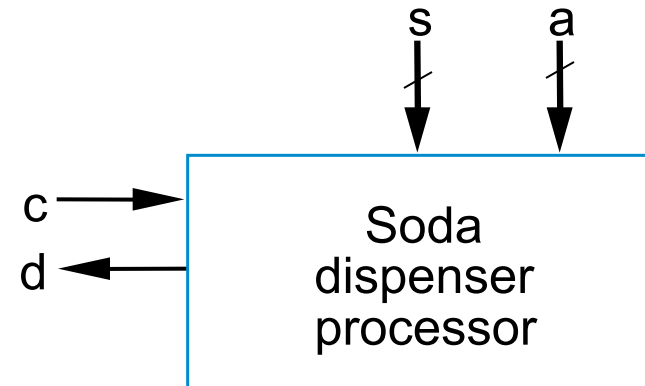
Processors:

- Programmable (microprocessor)
- Custom



High-Level State Machines (HLSMs)

- Some behaviors too complex for equations, truth tables, or FSMs
- Ex: Soda dispenser
 - c : bit input, 1 when coin deposited
 - a : 8-bit input having value of deposited coin
 - s : 8-bit input having cost of a soda
 - d : bit output, processor sets to 1 when total value of deposited coins equals or exceeds cost of a soda
- FSM can't represent...
 - 8-bit input/output
 - Storage of current total
 - Addition (e.g., $25 + 10$)



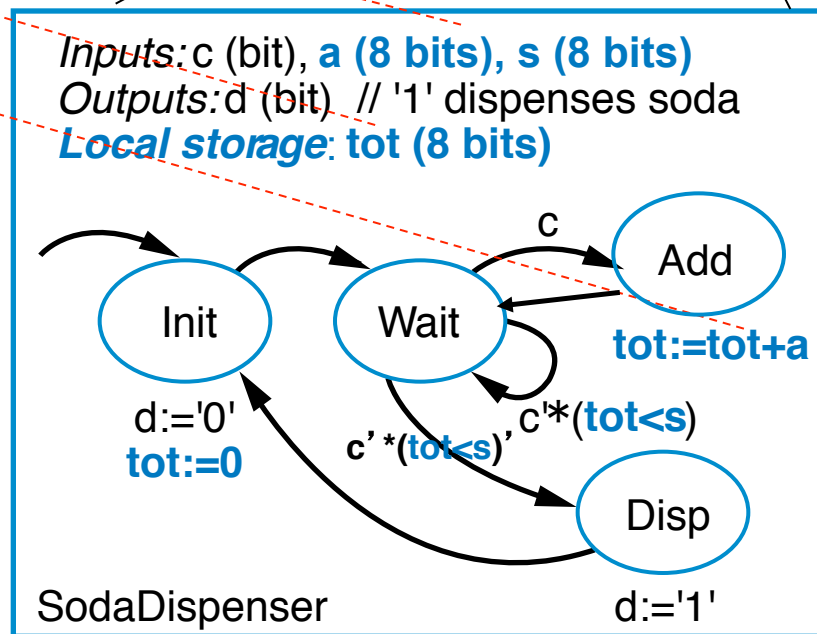
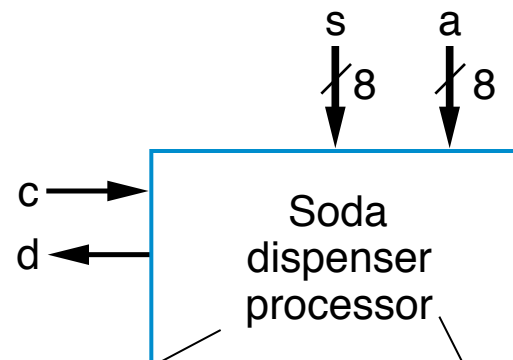
HLSMs

- High-level state machine (HLSM) extends FSM with:

- Multi-bit input/output
- Local storage
- Arithmetic operations

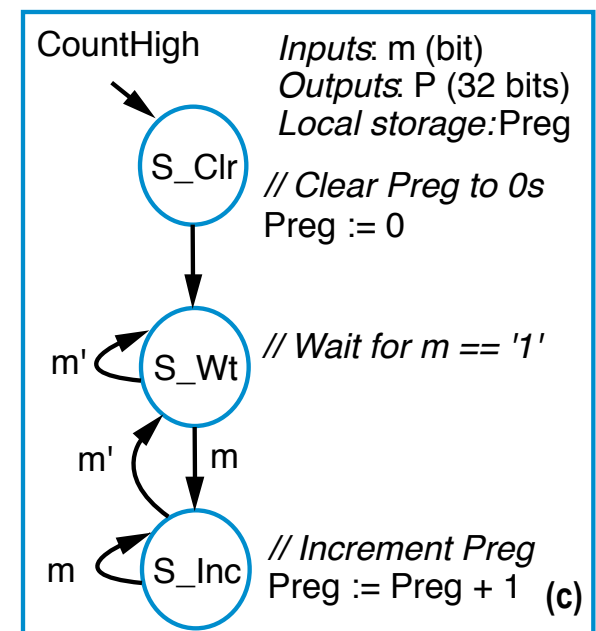
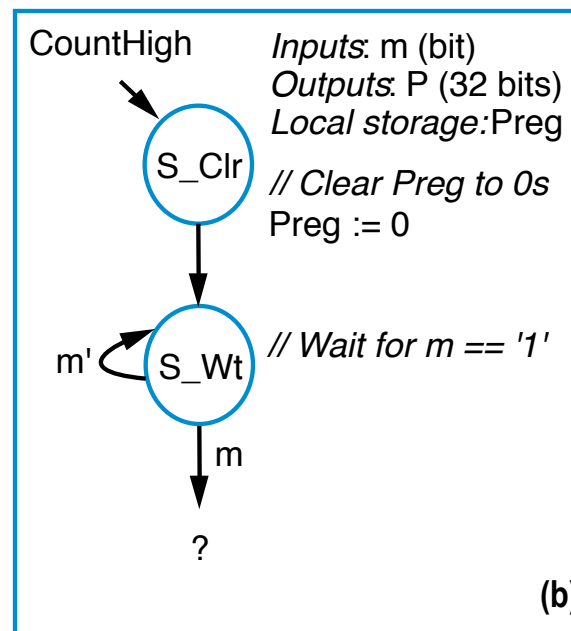
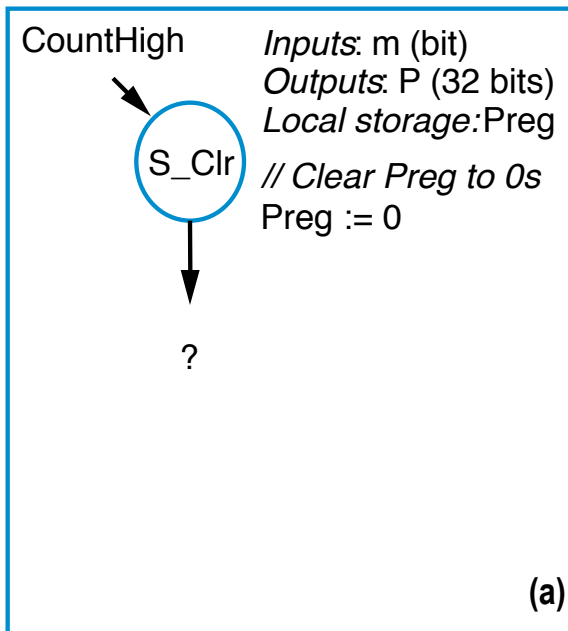
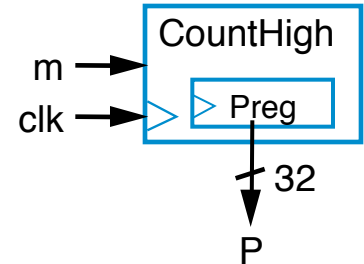
- Conventions

- Numbers:
 - Single-bit: '0' (single quotes)
 - Integer: 0 (no quotes)
 - Multi-bit: "0000" (double quotes)
- == for equal, := for assignment
- Multi-bit outputs *must* be registered via local storage
- // precedes a comment

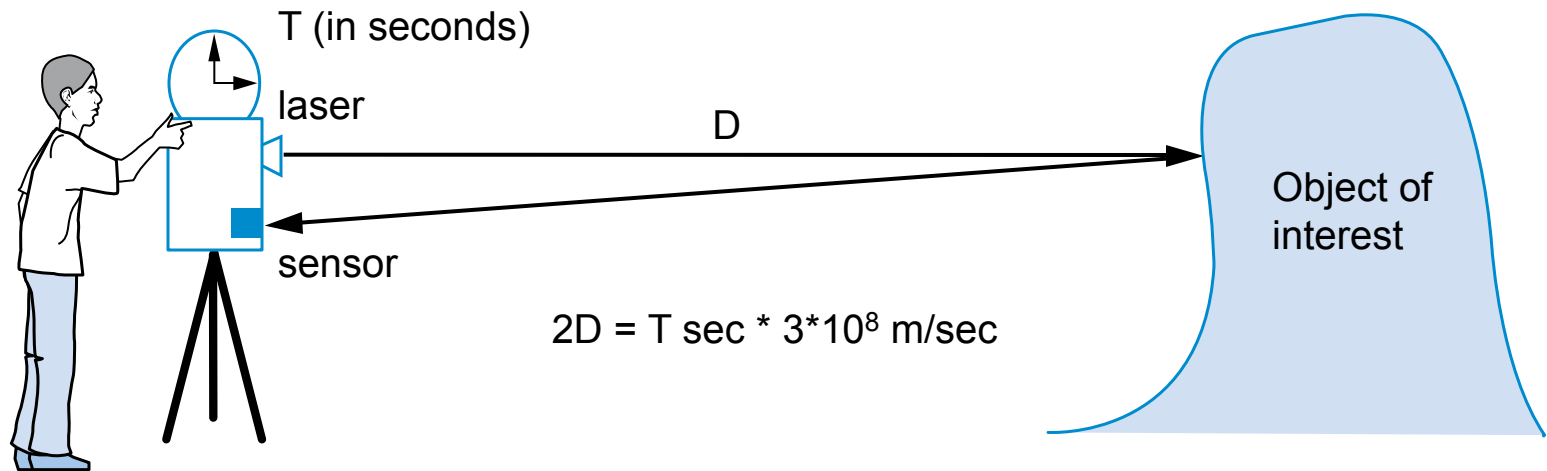


Ex: Cycles-High Counter

- P = total number (in binary) of cycles that m is 1
- Capture behavior as HLSM
 - Preg required (multibit outputs must be registered)
 - Use to hold count



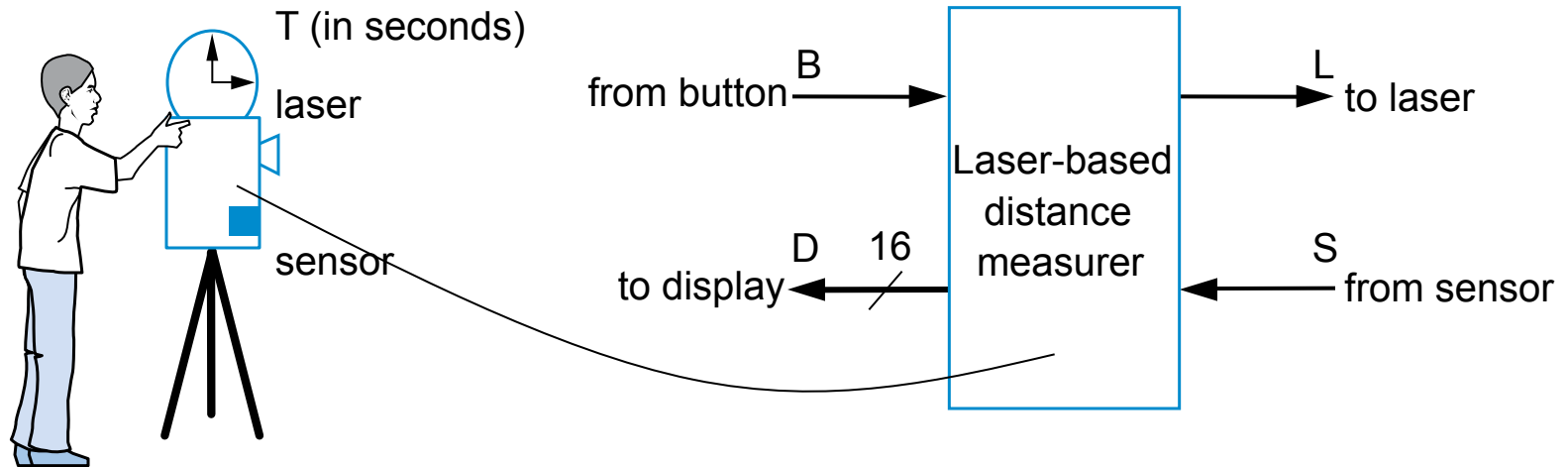
Example: Laser-Based Distance Measurer



- Laser-based distance measurement – pulse laser, measure time T to sense reflection
 - Laser light travels at speed of light, $3 * 10^8 \text{ m/sec}$
 - Distance is thus $D = (T \text{ sec} * 3 * 10^8 \text{ m/sec}) / 2$



Example: Laser-Based Distance Measurer

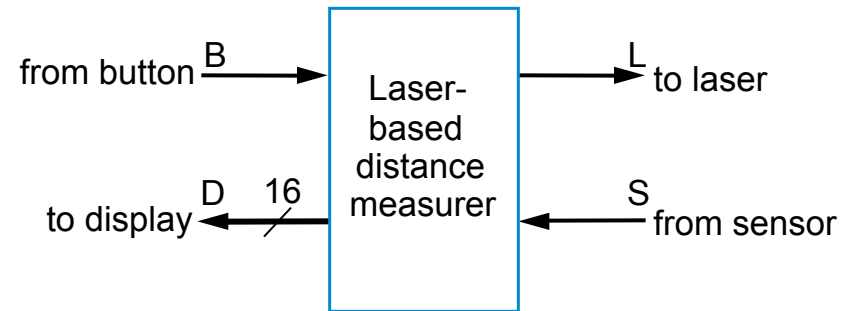
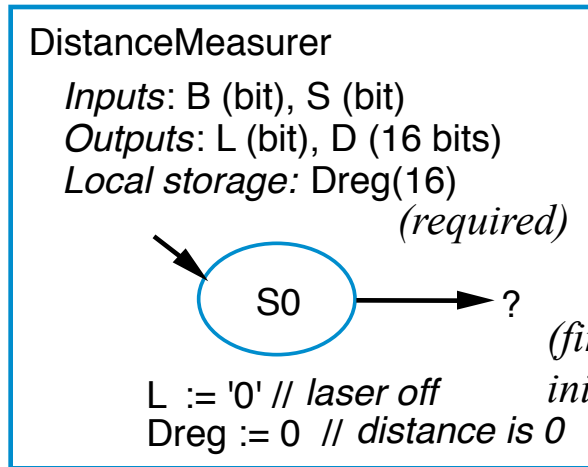


- Inputs/outputs

- B : bit input, from button, to begin measurement
- L : bit output, activates laser
- S : bit input, senses laser reflection
- D : 16-bit output, to display computed distance



Example: Laser-Based Distance Measurer

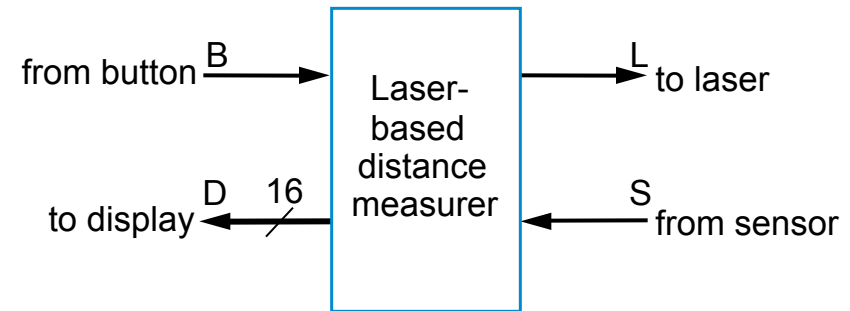
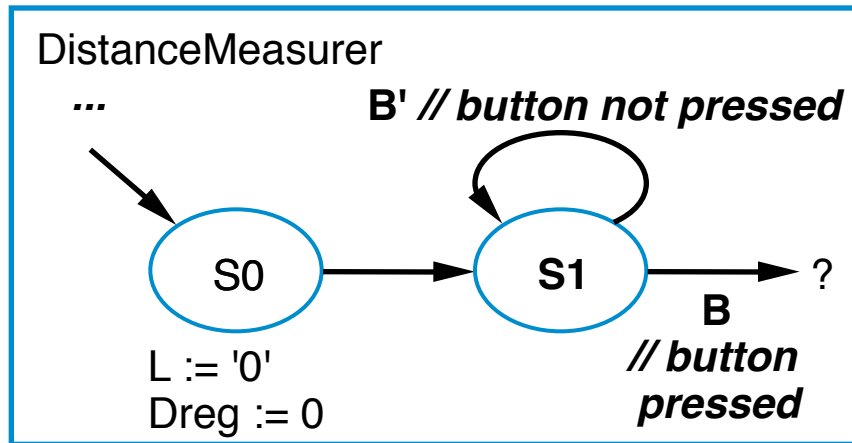


- Declare inputs, outputs, and local storage
 - Dreg required for multi-bit output
- Create initial state, name it **S0**
 - Initialize laser to off (L:='0')
 - Initialize displayed distance to 0 (Dreg:=0)

*Recall: '0' means single bit,
 0 means integer*



Example: Laser-Based Distance Measurer

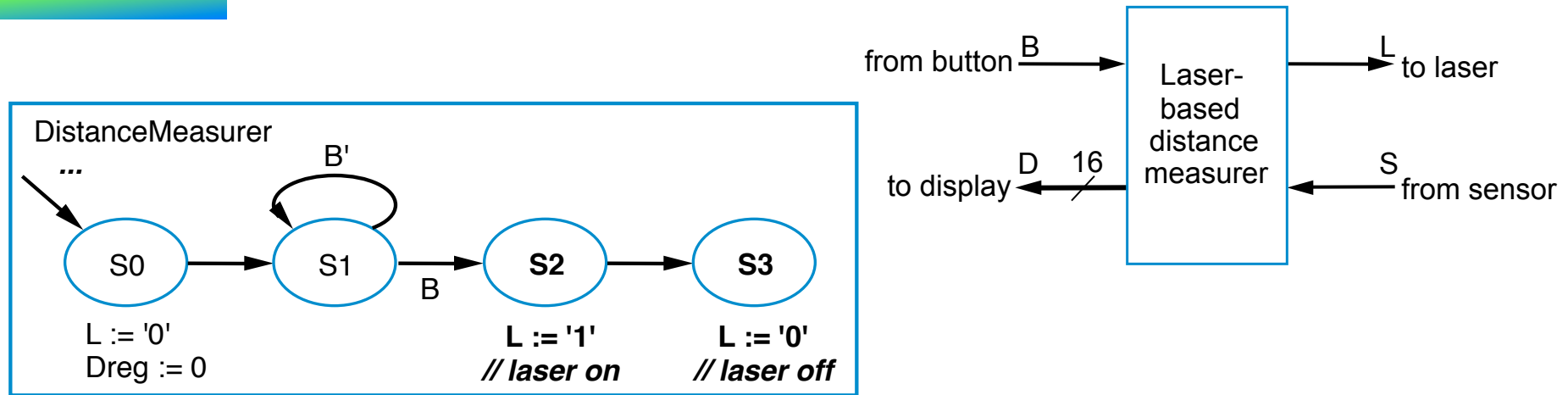


- Add another state, **S1**, that waits for a button press
 - B' – stay in **S1**, keep waiting
 - B – go to a new state **S2**

Q: What should S2 do? **A: Turn on the laser**



Example: Laser-Based Distance Measurer



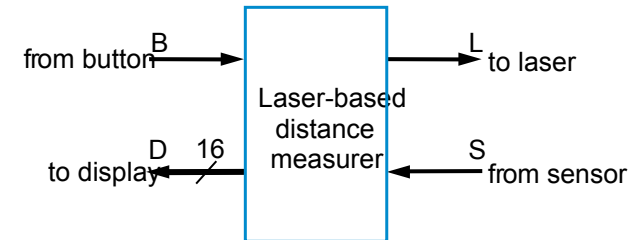
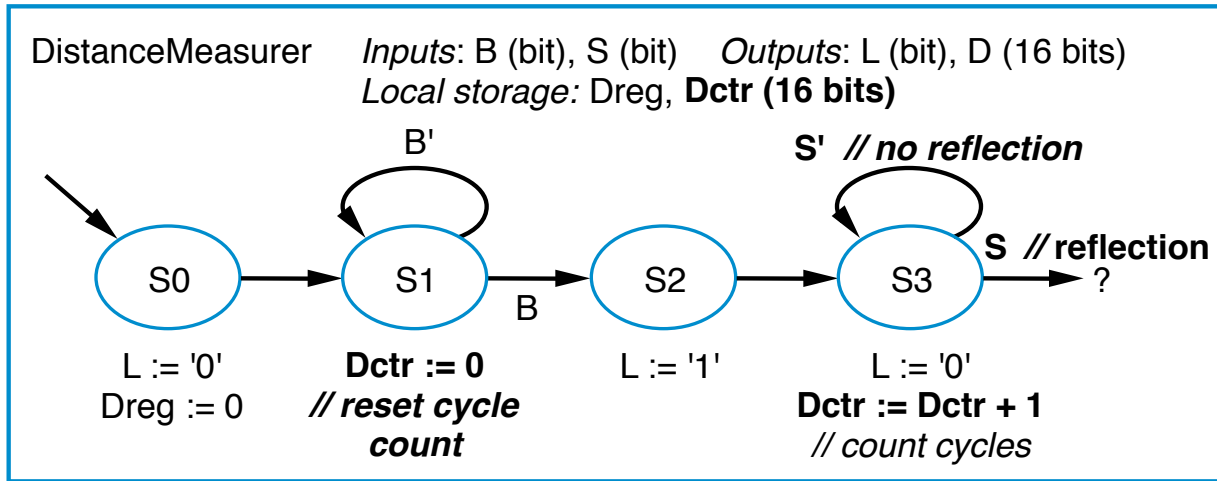
- Add a state **S2** that turns on the laser ($L:='1'$)
- Then turn off laser ($L:='0'$) in a state **S3**

Q: What do next? A: Start timer, wait to sense reflection

a



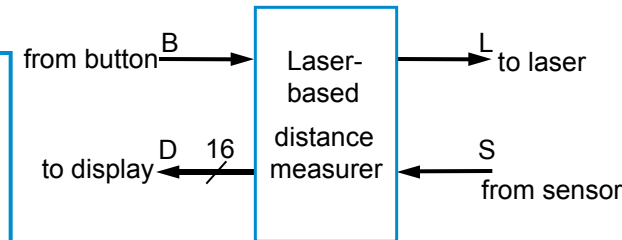
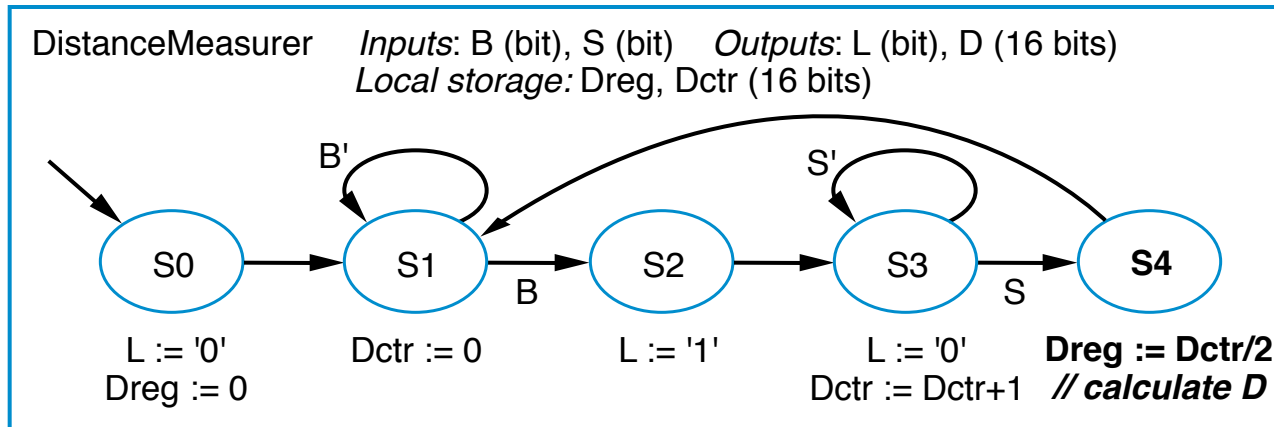
Example: Laser-Based Distance Measurer



- Stay in **S3** until sense reflection (S)
- To measure time, count cycles while in **S3**
 - To count, declare local storage *Dctr*
 - Initialize *Dctr* to 0 in **S1**. In **S2** would have been O.K. too.
 - Don't forget to initialize local storage—common mistake
 - Increment *Dctr* each cycle in **S3**



Example: Laser-Based Distance Measurer

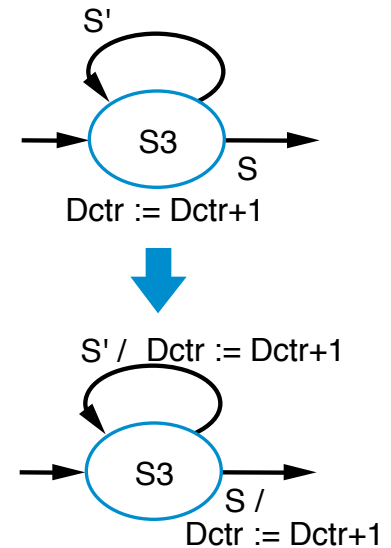


- Once reflection detected (S), go to new state **S4**
 - Calculate distance
 - Assuming clock frequency is 3×10^8 , $Dctr$ holds number of meters, so $Dreg := Dctr / 2$
- After **S4**, go back to **S1** to wait for button again

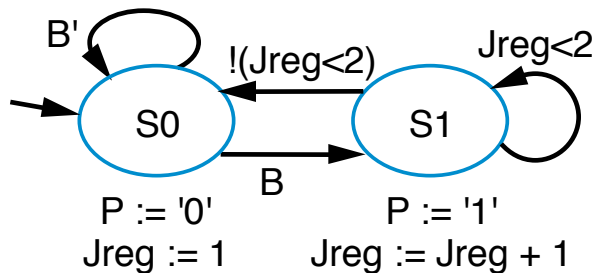


HLSM Actions: Updates Occur Next Clock Cycle

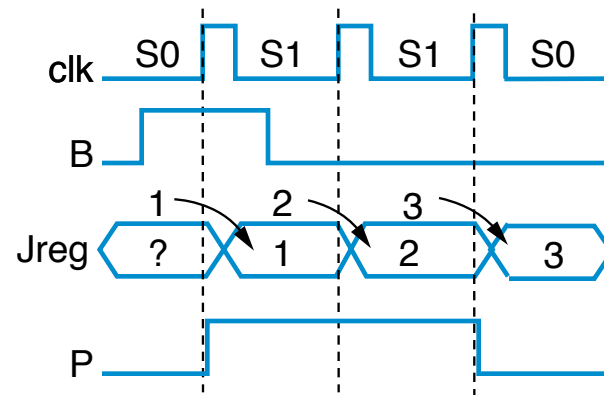
- Local storage updated on clock edges only
 - Enter state on clock edge
 - Storage writes in that state occur on *next* clock edge
 - Can think of as occurring on outgoing transitions
- Thus**, transition conditions use the OLD value, not the newly-written value
 - Example:



Inputs: B (bit)
 Outputs: P (bit) // if B, 2 cycles high
 Local storage: Jreg (8 bits)



(a)

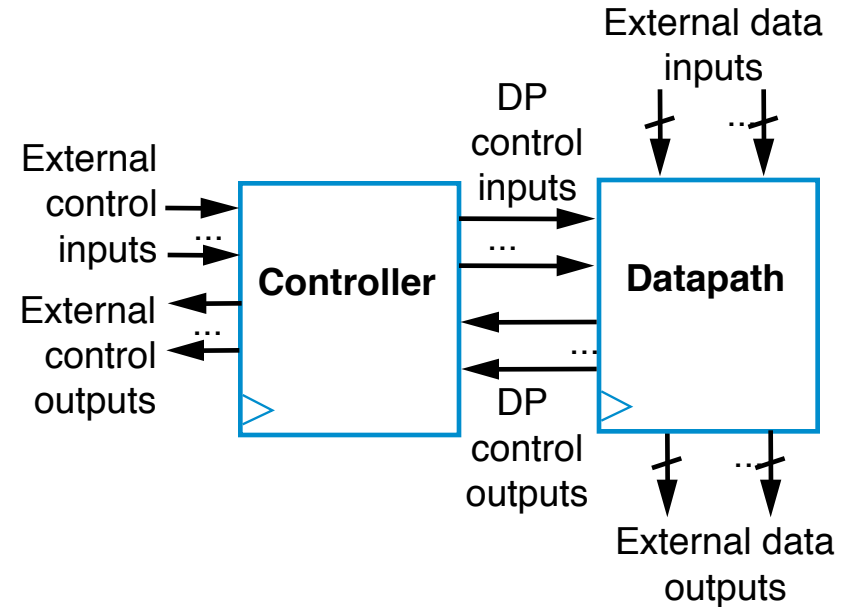


(b)

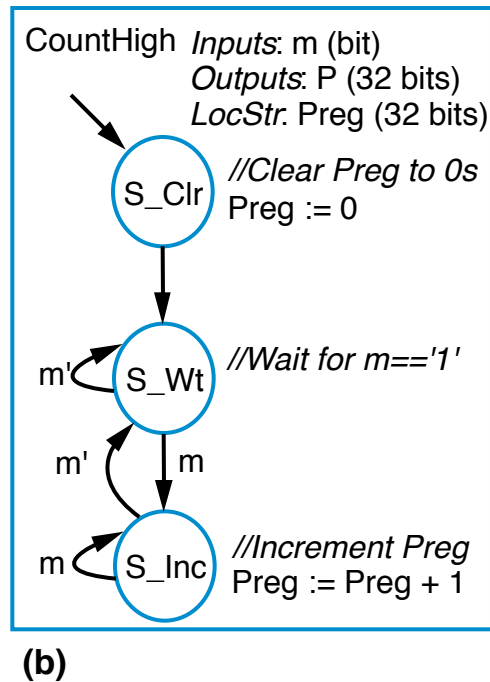
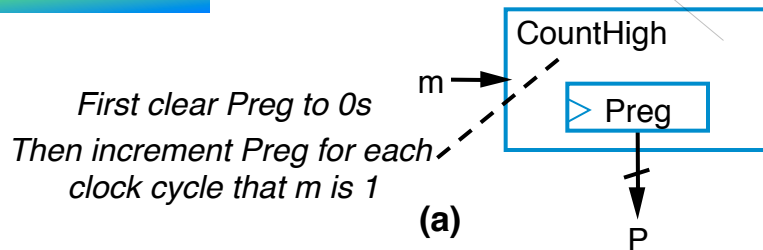


RTL Design Process

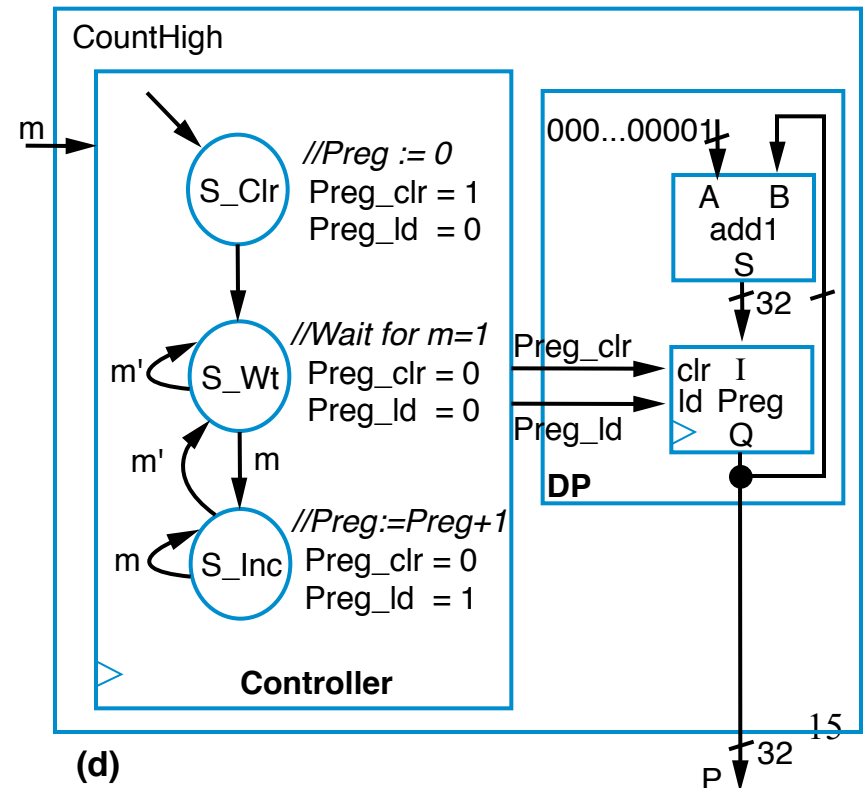
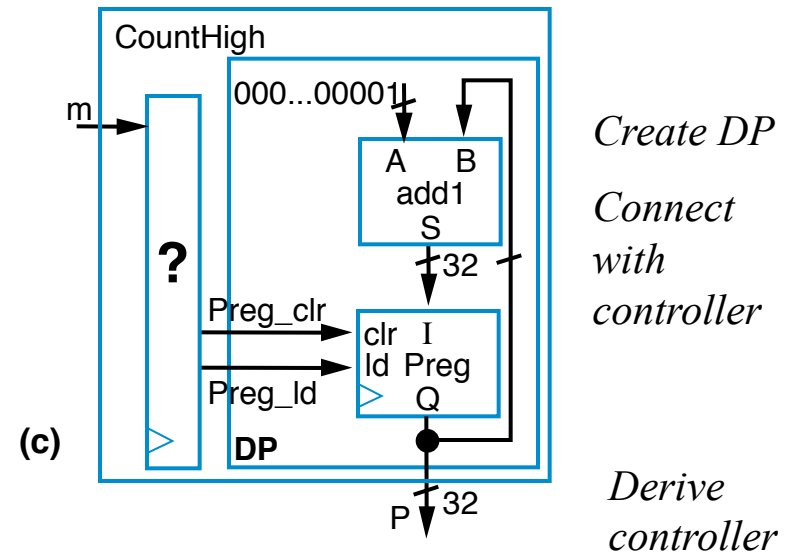
- Capture behavior
- Convert to circuit
 - Need target architecture
 - Datapath capable of HLSM's data operations
 - Controller to control datapath



Ctrl/DP Example for Earlier Cycles-High Counter



We created this HLSM earlier



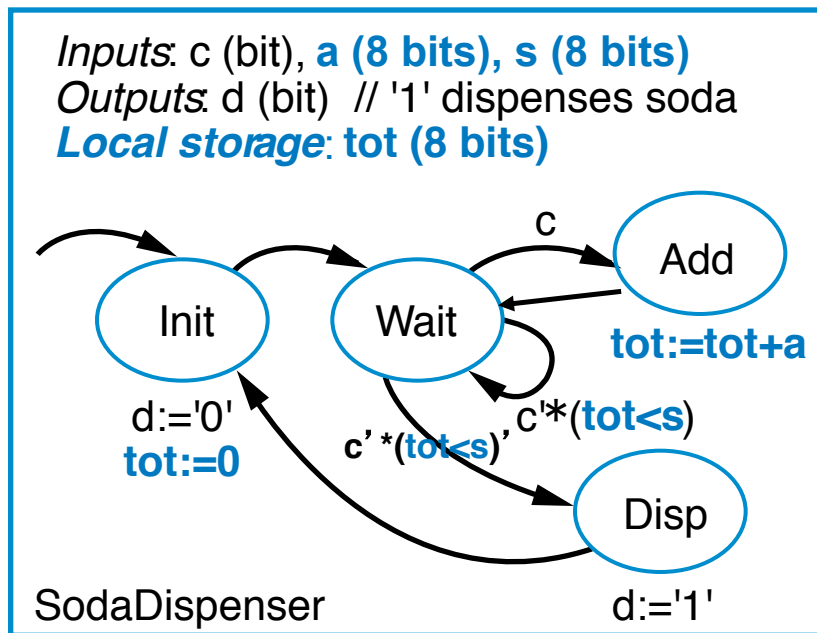
RTL Design Process

	Step	Description
Step 1: Capture behavior	<i>Capture a high-level state machine</i>	Describe the system's desired behavior as a high-level state machine. The state machine consists of states and transitions. The state machine is "high-level" because the transition conditions and the state actions are more than just Boolean operations on single-bit inputs and outputs.
	2A <i>Create a datapath</i>	Create a datapath to carry out the data operations of the high-level state machine.
Step 2: Convert to circuit	2B <i>Connect the datapath to a controller</i>	Connect the datapath to a controller block. Connect external control inputs and outputs to the controller block.
	2C <i>Derive the controller's FSM</i>	Convert the high-level state machine to a finite-state machine (FSM) for the controller, by replacing data operations with setting and reading of control signals to and from the datapath.

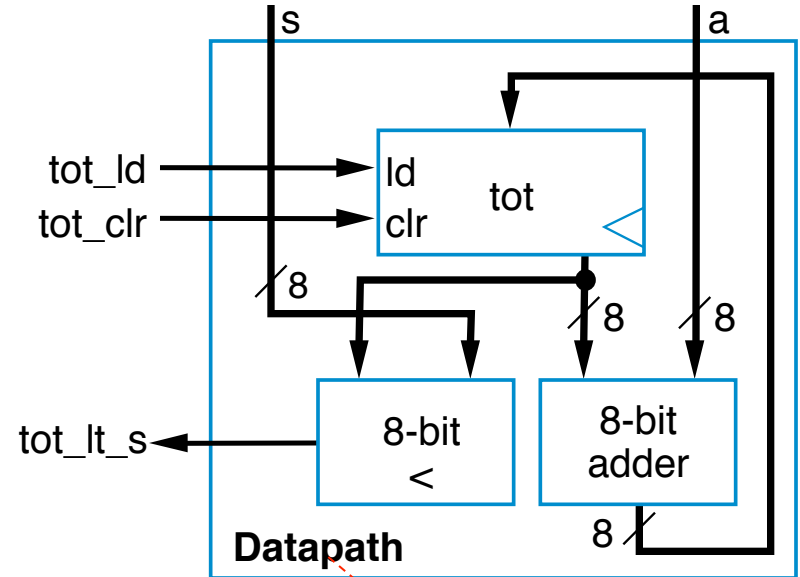


Example: Soda Dispenser from Earlier

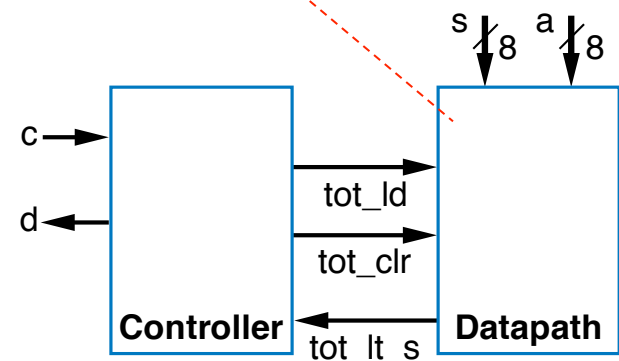
- Quick overview example.
More details of each step to come.



Step 1



Step 2A

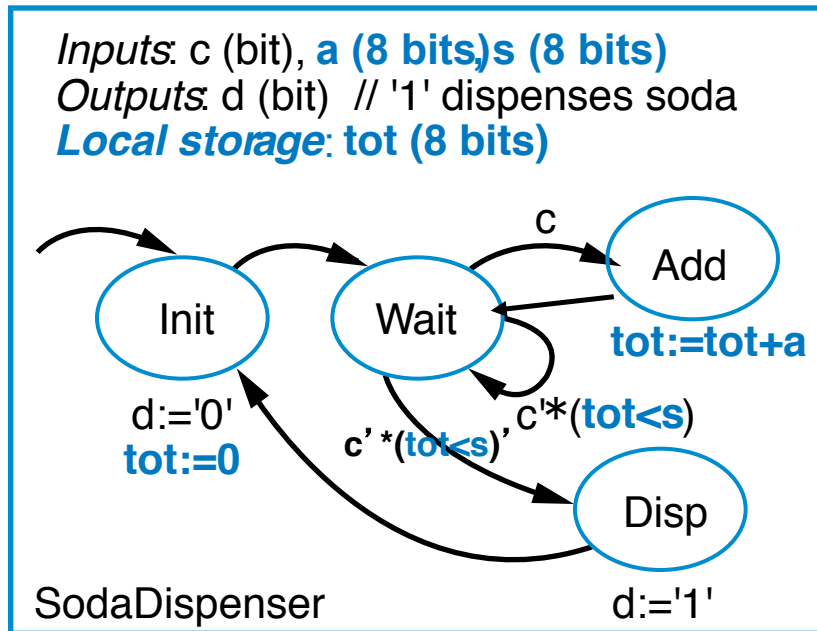


Step 2B

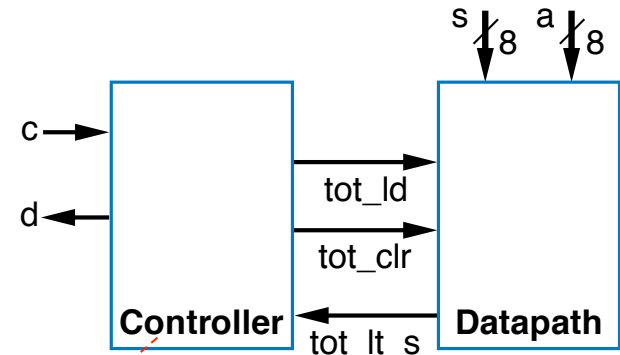


Example: Soda Dispenser

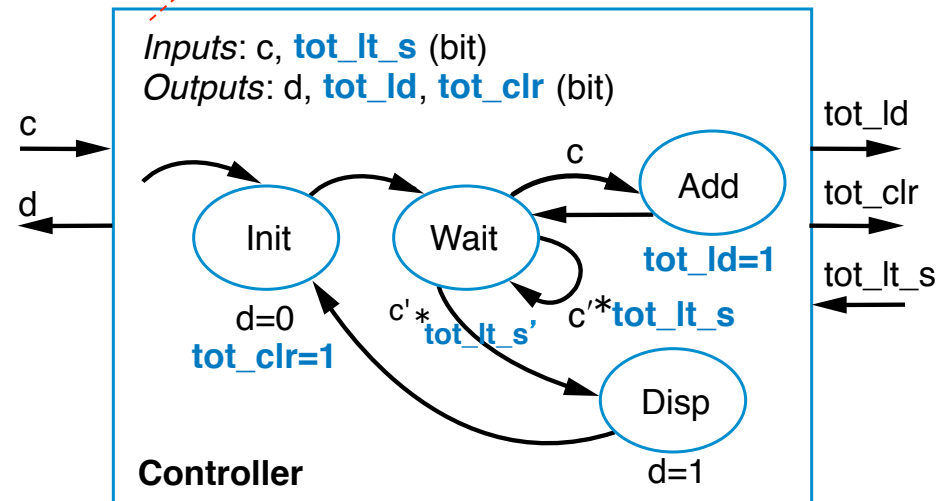
- Quick overview example.
More details of each step to come.



Step 1



Step 2B



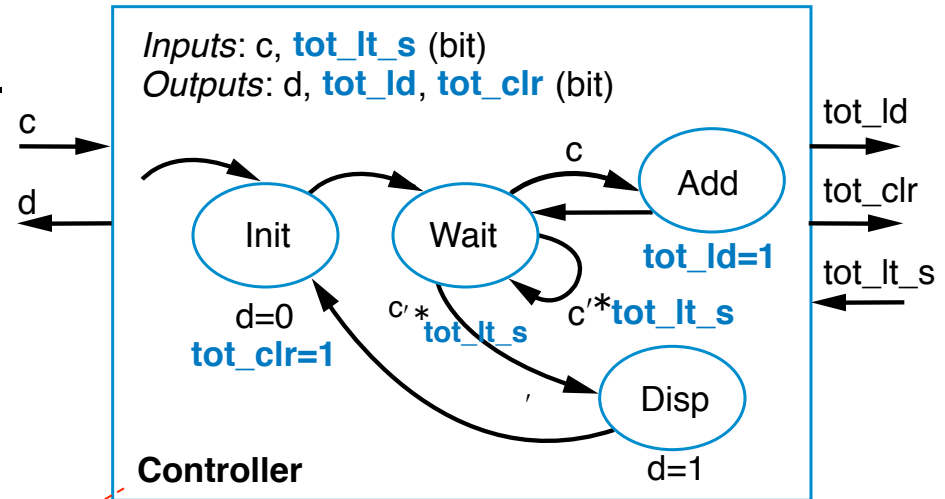
Step 2C



Example: Soda Dispenser

- Quick overview example.
More details of each step to come.

	s1	s0	c	tot_lt_s	n1	n0	d	tot_ld	tot_clr
Init	0	0	0	0	0	1	0	0	1
	0	0	0	1	0	1	0	0	1
	0	0	1	0	0	1	0	0	1
	0	0	1	1	0	1	0	0	1
Wait	0	1	0	0	1	1	0	0	0
	0	1	0	1	0	1	0	0	0
	0	1	1	0	1	0	0	0	0
	0	1	1	1	1	0	0	0	0
Add	1	0	0	0	0	1	0	1	0
			...			...			
Disp	1	1	0	0	0	0	1	0	0
			...			...			



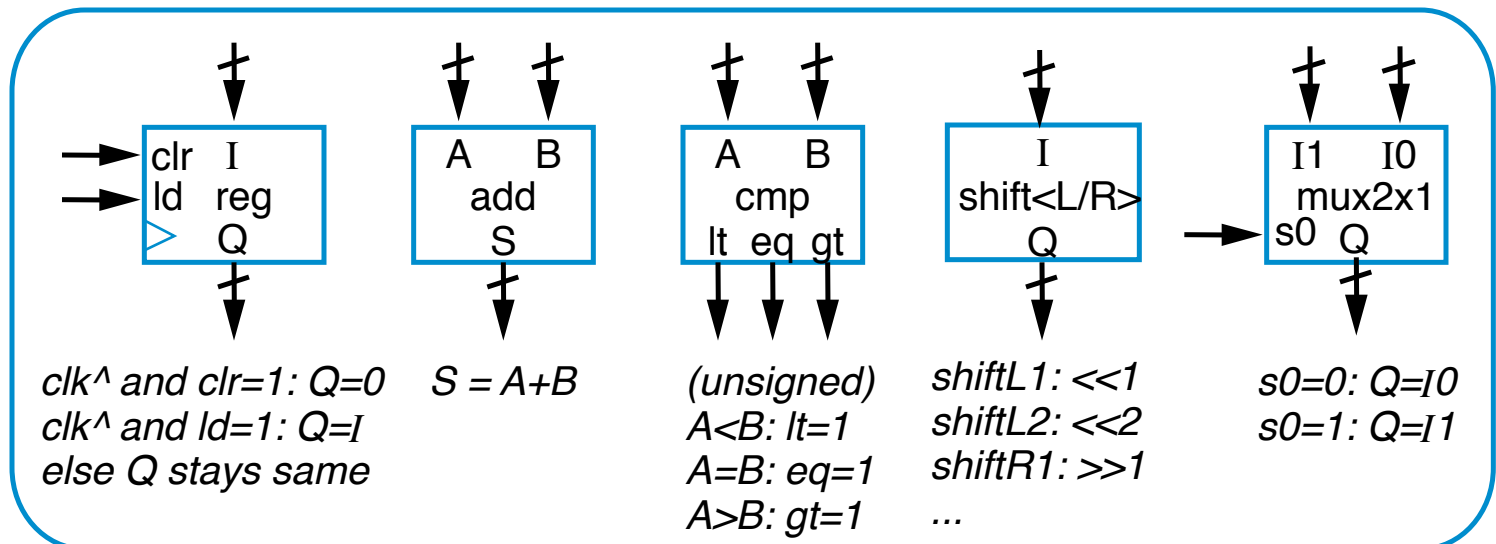
Step 2C

Use controller design process
(Ch3) to complete the design

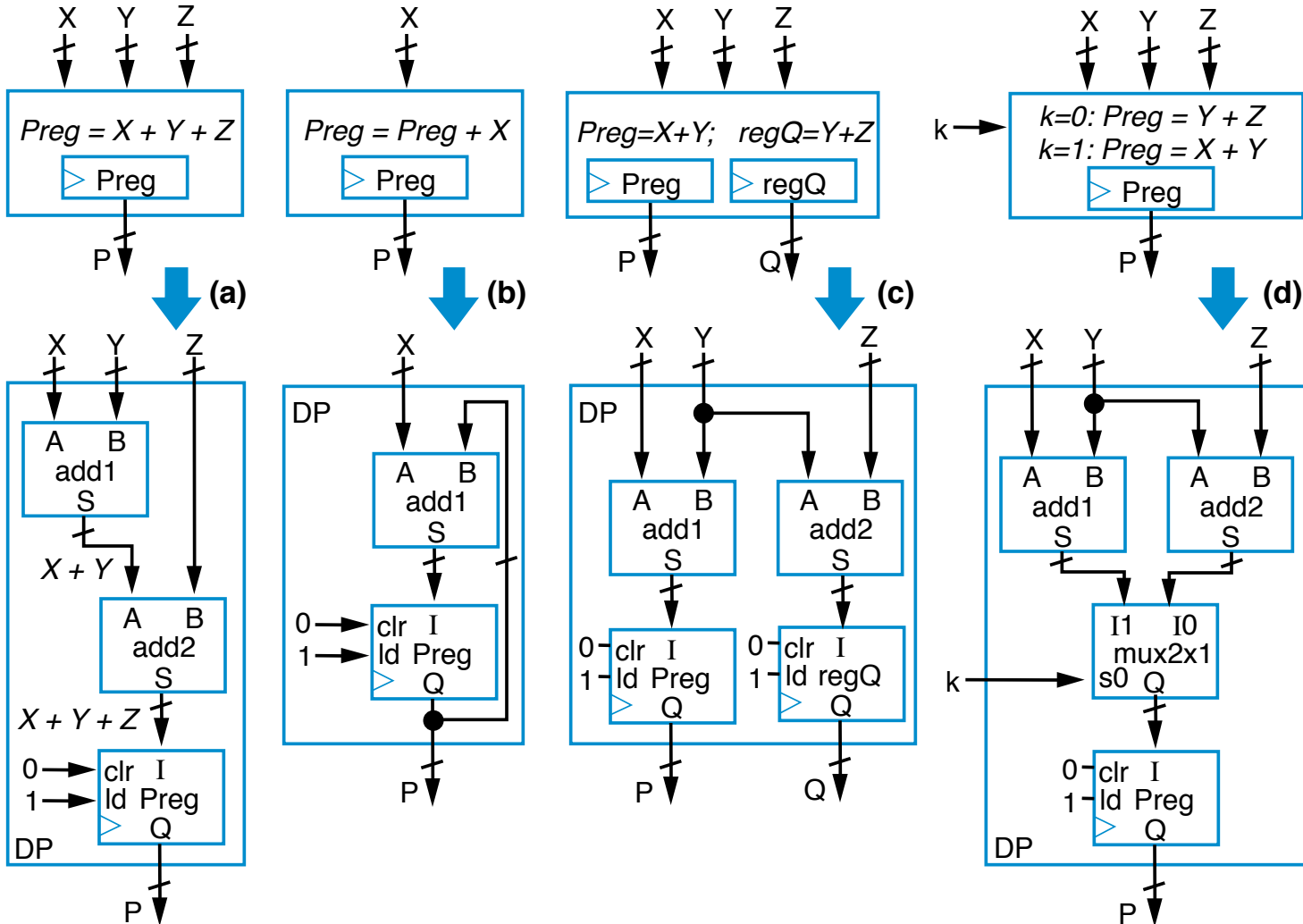


RTL Design Process—Step 2A: Create a datapath

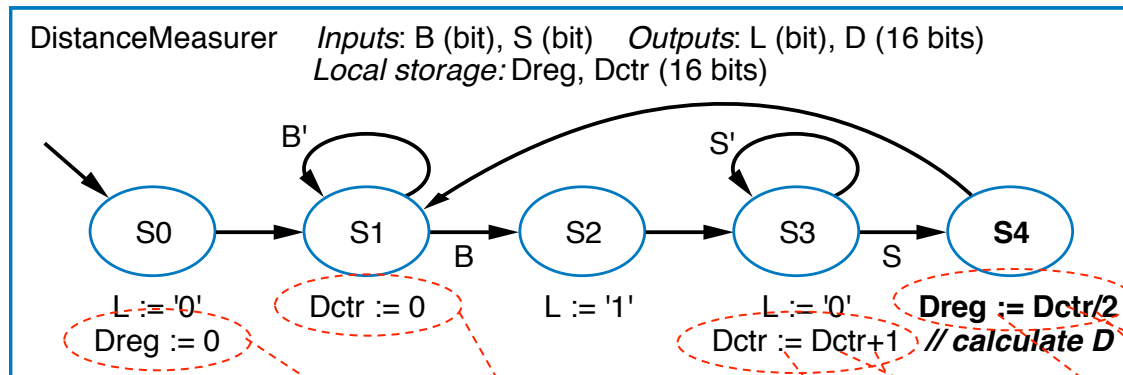
- Sub-steps
 - HLSM data inputs/outputs → Datapath inputs/outputs.
 - HLSM local storage item → Instantiated register
 - "Instantiate": Add new component ("instance") to design
 - Each HLSM state action and transition condition data computation → Datapath components and connections
 - Also instantiate multiplexors as needed
- Need component library from which to choose



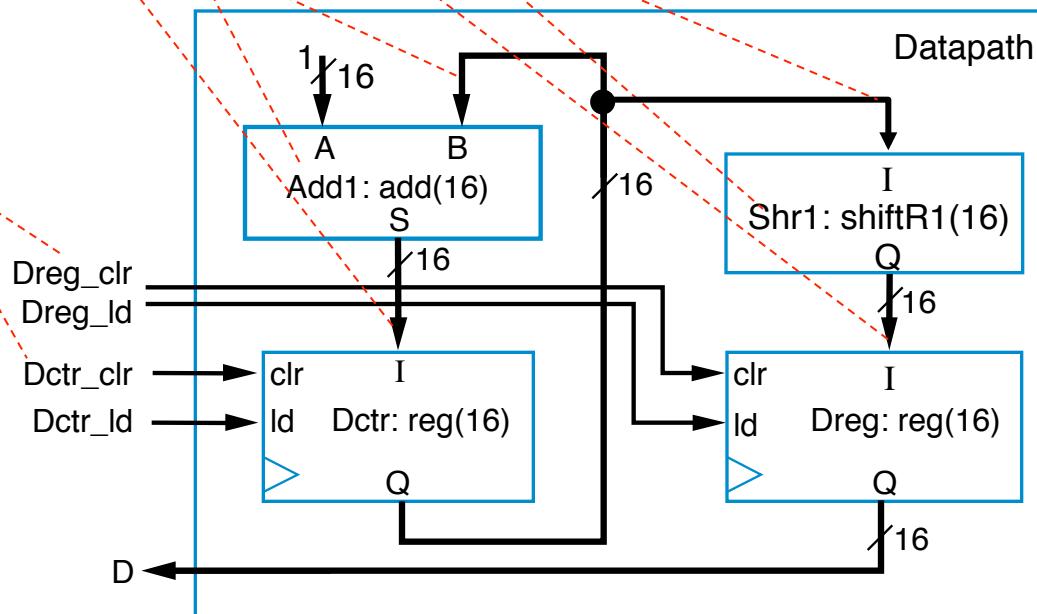
Step 2A: Create a Datapath—Simple Examples



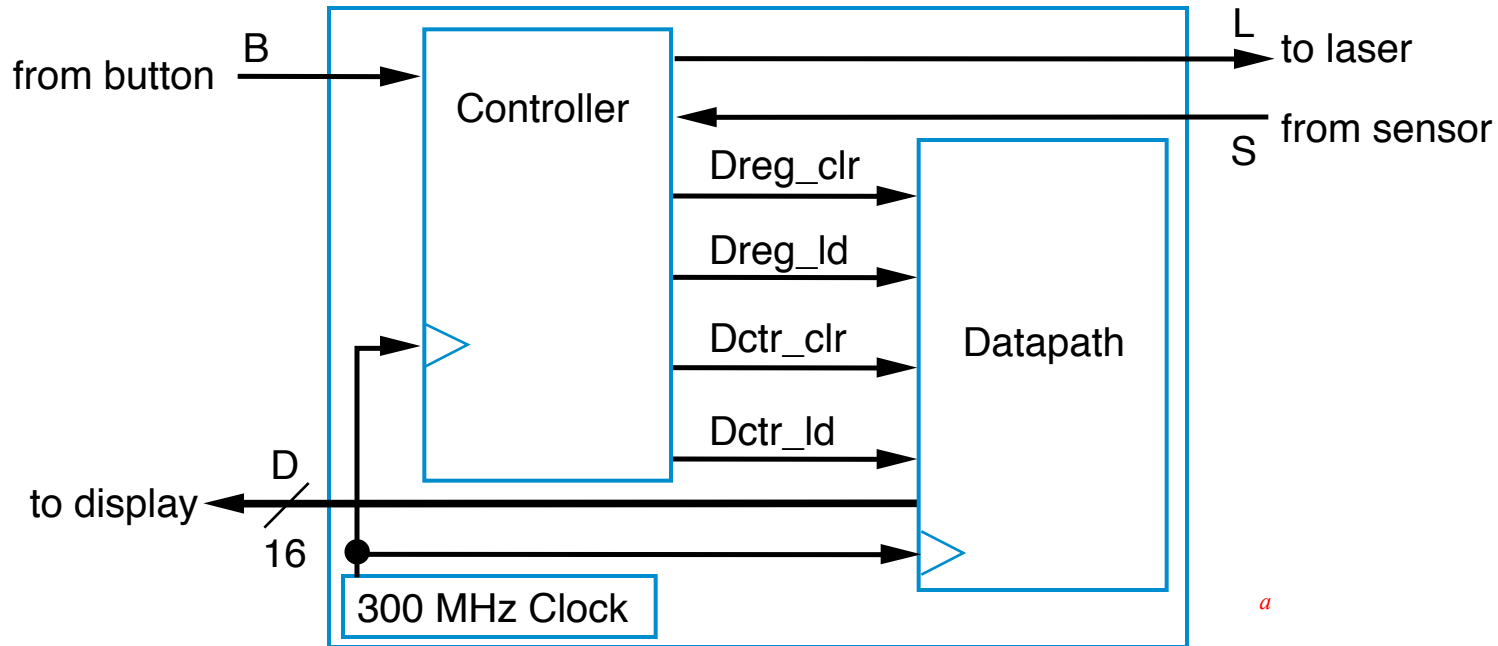
Laser-Based Distance Measurer—Step 2A: Create a Datapath



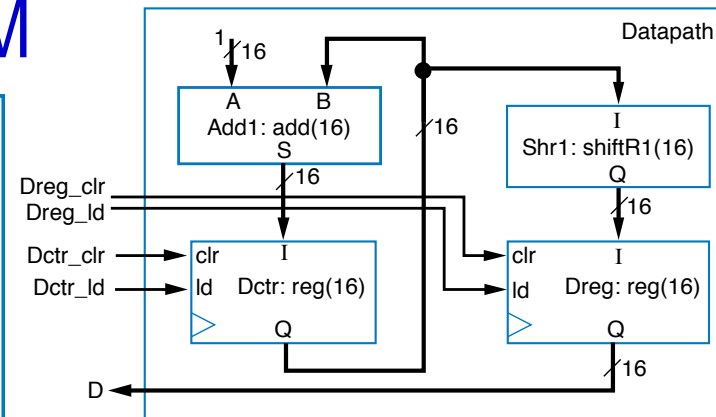
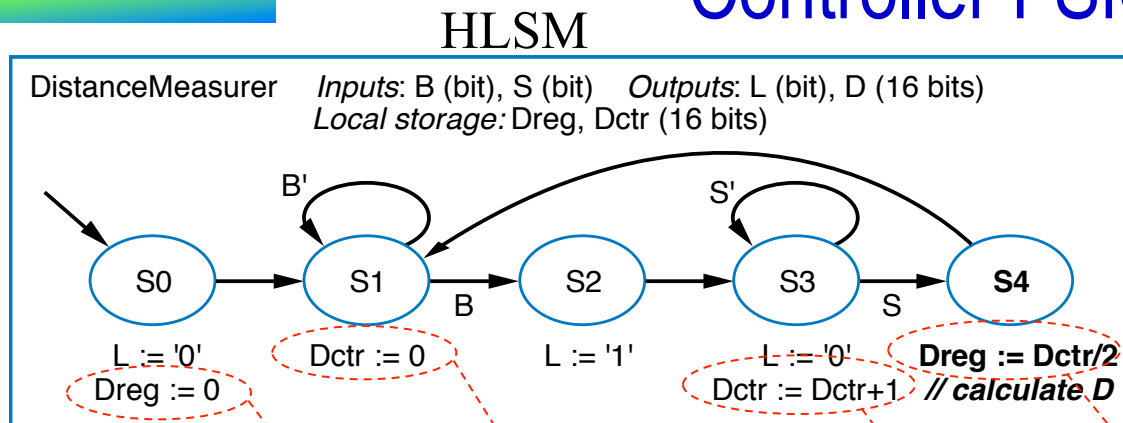
- HLSM data I/O $\rightarrow$ DP I/O
- HLSM local storage $\rightarrow$ reg
- HLSM state action and transition condition data computation $\rightarrow$ Datapath components and connections



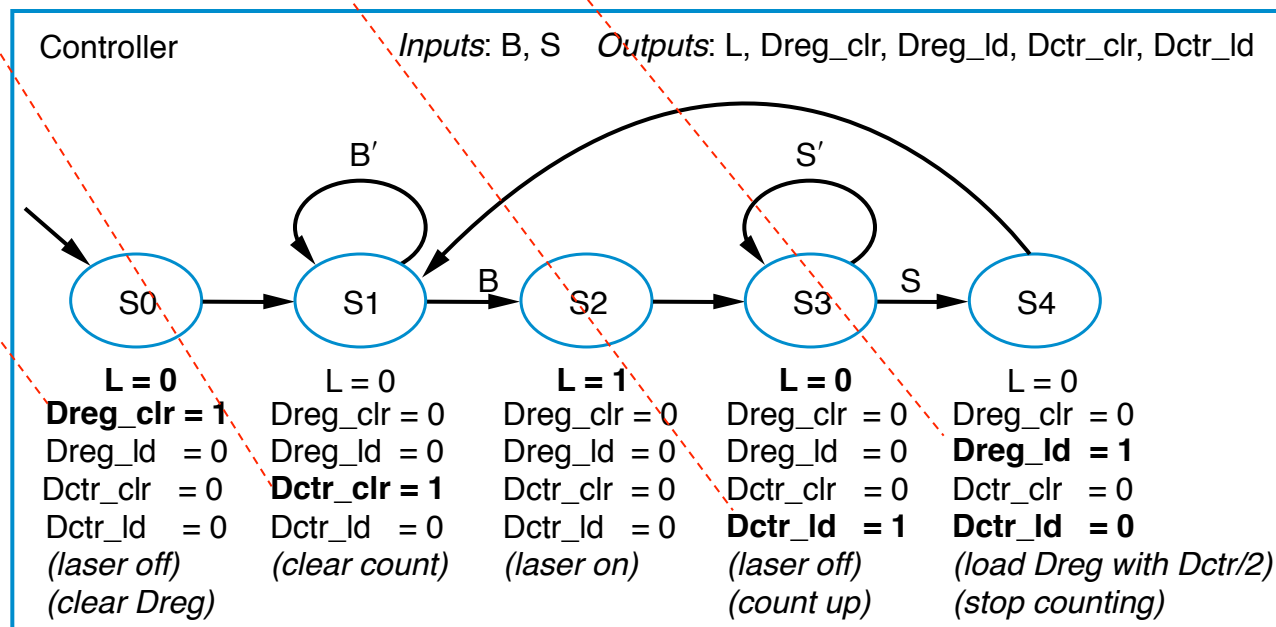
Laser-Based Distance Measurer—Step 2B: Connecting the Datapath to a Controller



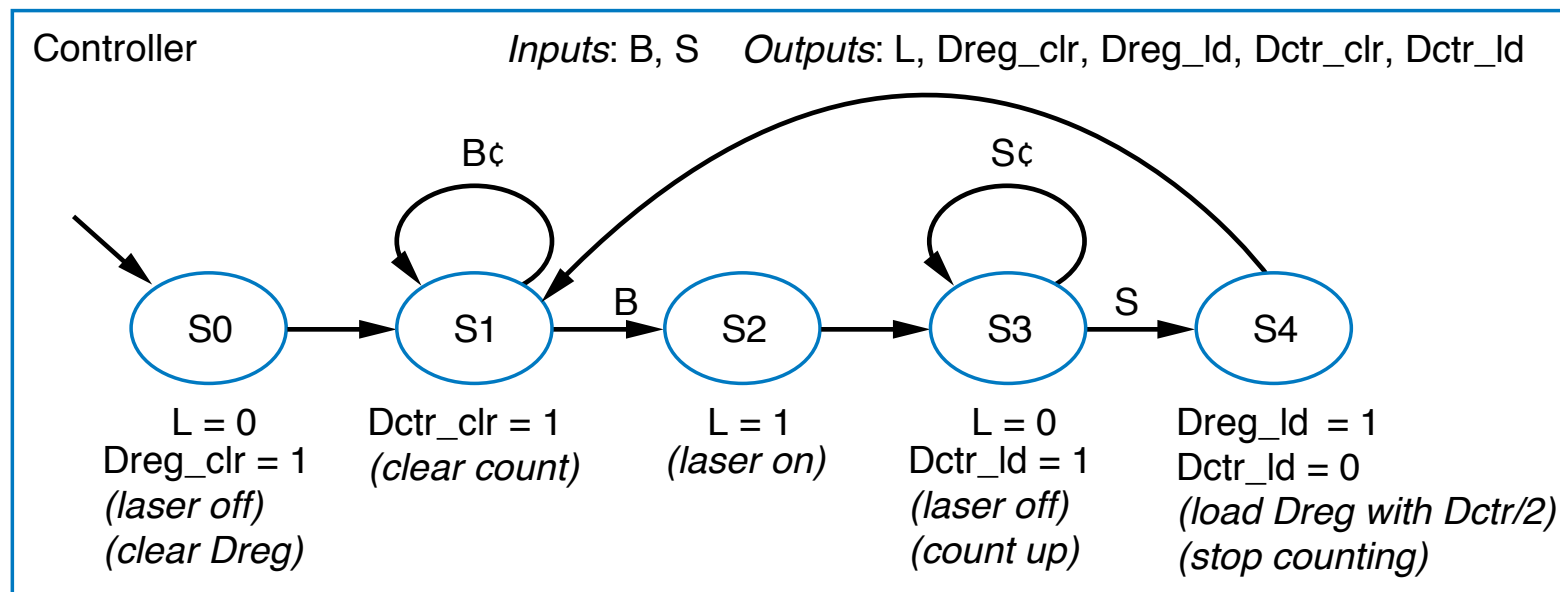
Laser-Based Distance Measurer—Step 2C: Derive the Controller FSM



- FSM has same states, transitions, and control I/O
- Achieve each HLSM data operation using datapath control signals in FSM



Laser-Based Distance Measurer—Step 2C: Derive the Controller FSM



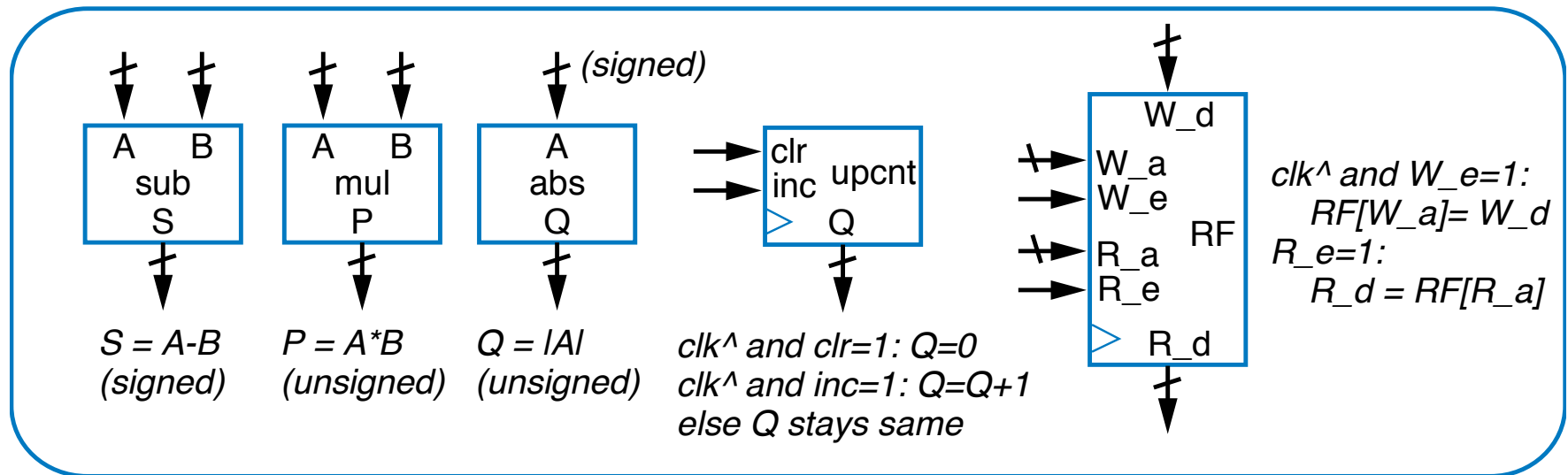
- Same FSM, using convention of unassigned outputs implicitly assigned 0

Some assignments to 0 still shown, due to their importance in understanding desired controller behavior



More RTL Design

- Additional datapath components

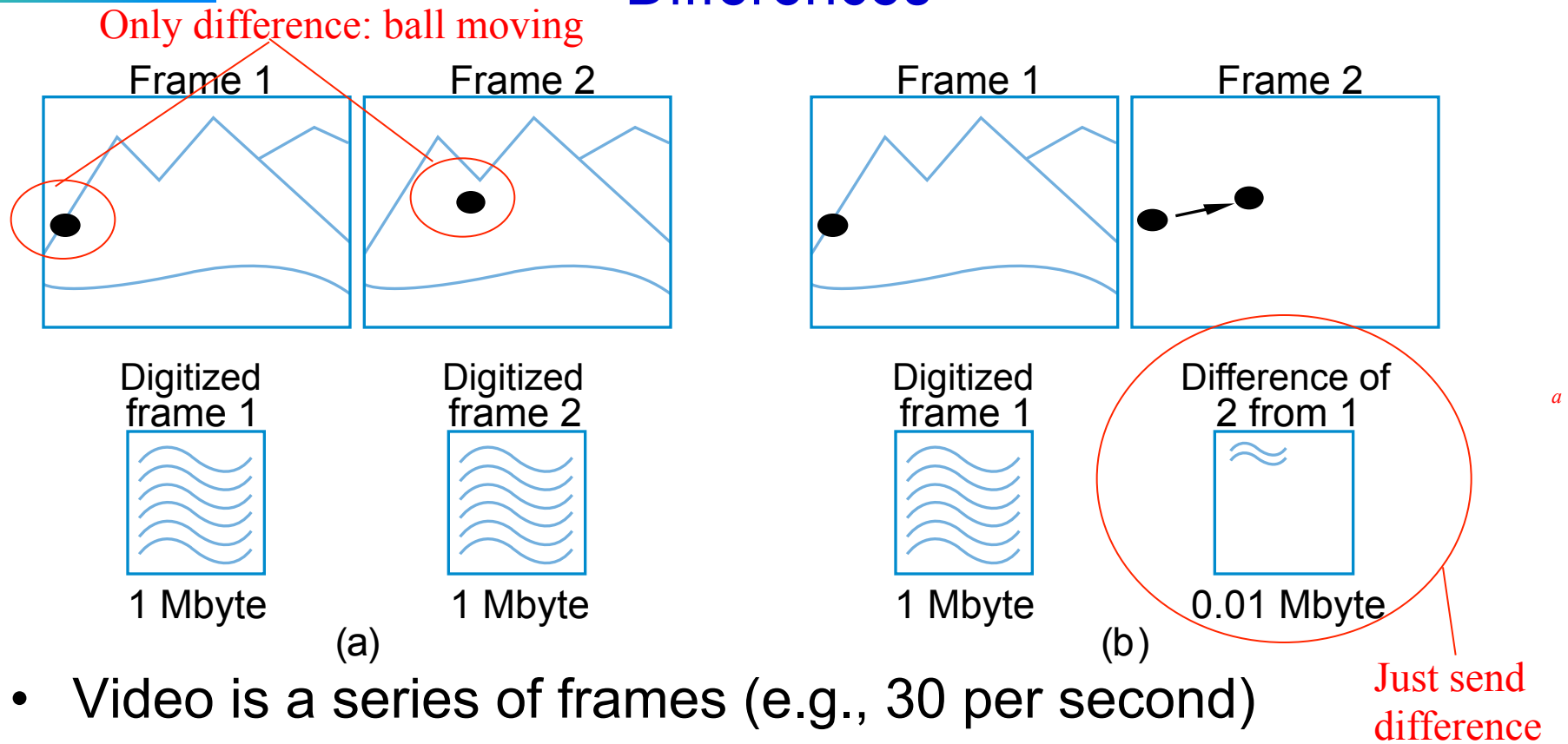


RTL Design Involving Register File or Memory

- HLASM *array*: Ordered list of items
 - Ex: Local storage: A[4](8-bit) – 4 8-bit items
 - Accessed using notation "A[i]", i is *index*
 - A[0] := 9; A[1] := 8; A[2] := 7; A[3] := 22
 - Array contents now: <9, 8, 7, 22>
 - X := A[1] will set X to 8
 - Note: First element's index is 0
- Array can be mapped to instantiated register file or memory



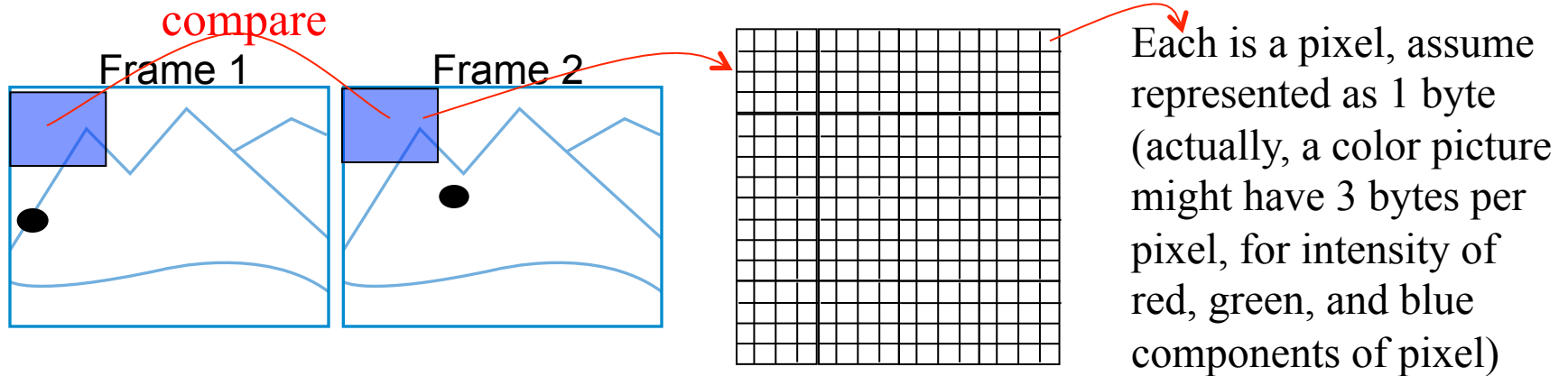
RTL Example: Video Compression – Sum of Absolute Differences



- Video is a series of frames (e.g., 30 per second)
- Most frames similar to previous frame
 - Compression idea: just send difference from previous frame



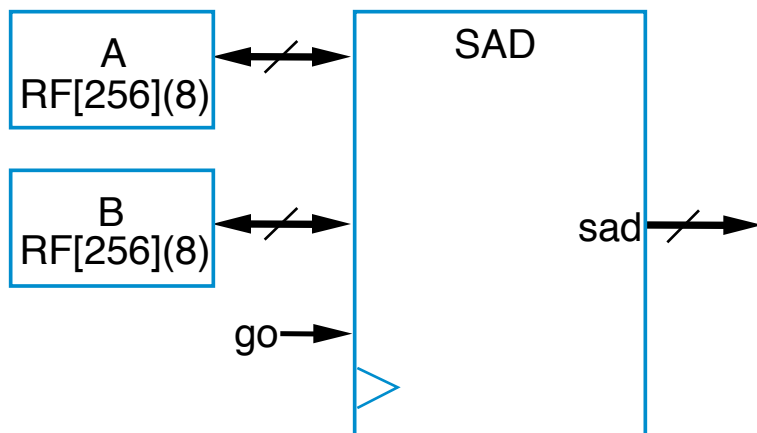
RTL Example: Video Compression – Sum of Absolute Differences



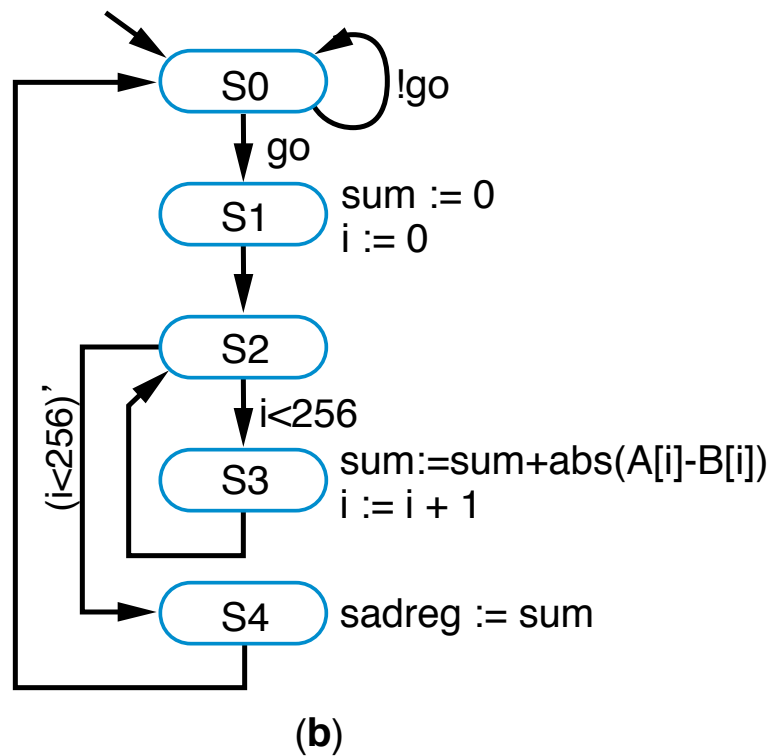
- Need to quickly determine whether two frames are similar enough to just send difference for second frame
 - Compare corresponding 16x16 “blocks”
 - Treat 16x16 block as 256-byte array
 - Compute the absolute value of the difference of each array item
 - Sum those differences – if above a threshold, send complete frame for second frame; if below, can use difference method (using another technique, not described)



Array Example: Video Compression—Sum-of-Absolute Differences



Inputs: A, B [256](8 bits); go (bit)
 Outputs: sad (32 bits)
 Local storage: sum, sadreg (32 bits); i (9 bits)

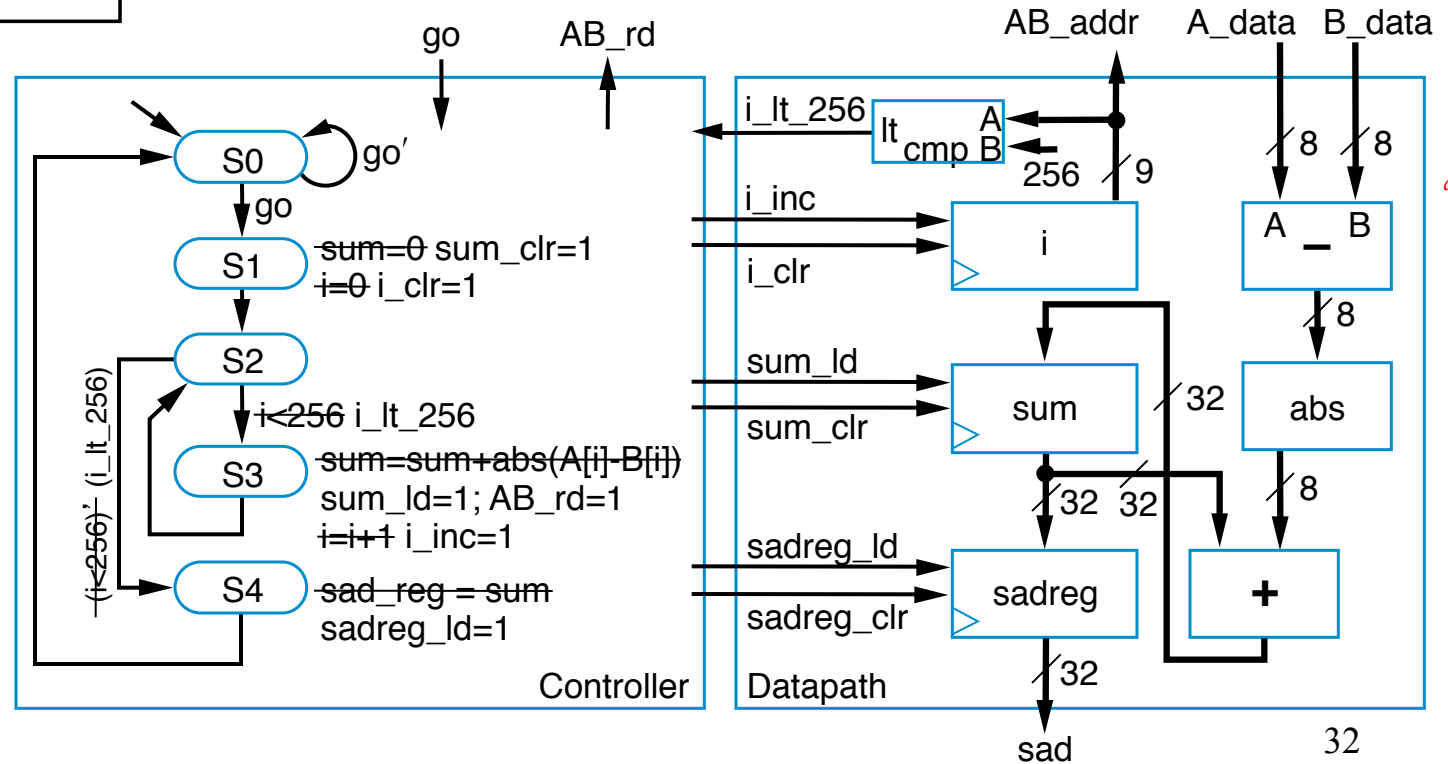
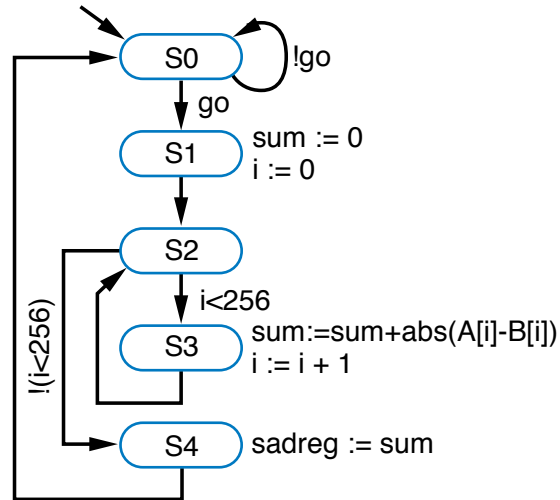


- **S0**: wait for *go*
- **S1**: initialize *sum* and *index*
- **S2**: check if done (*(i < 256)'*)
- **S3**: add difference to *sum*, increment index
- **S4**: done, write to output *sad_reg*



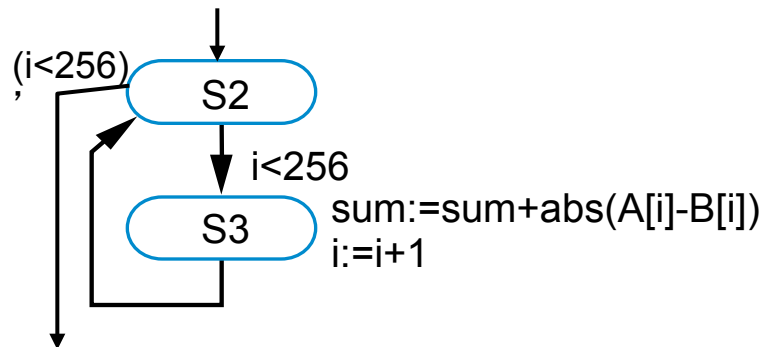
Inputs: A, B [256](8 bits); go (bit)
 Outputs: sad (32 bits)
 Local storage: sum, sadreg (32 bits); i (9 bits)

Array Example: Video Compression—Sum-of-Absolute Differences



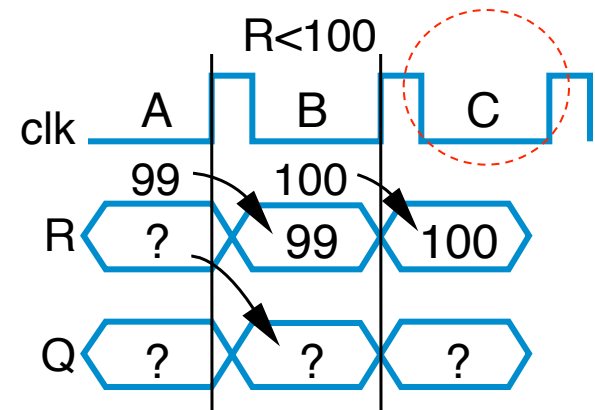
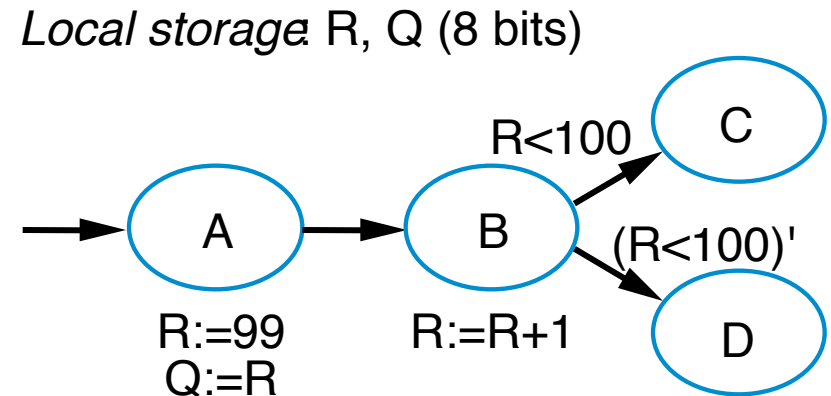
Circuit vs. Microprocessor

- Circuit: Two states (**S2** & **S3**) for each i , 256 i 's $\rightarrow$ 512 clock cycles
- Microprocessor: Loop (*for* $i = 1$ to 256), but for each i , must move memory to local registers, subtract, compute absolute value, add to sum, increment i – say 6 cycles per array item $\rightarrow 256 \cdot 6 = 1536$ cycles
- Circuit is about 3 *times* (300%) faster (assuming equal cycle lengths)
- Later, we'll see how to build SAD circuit that is *much* faster



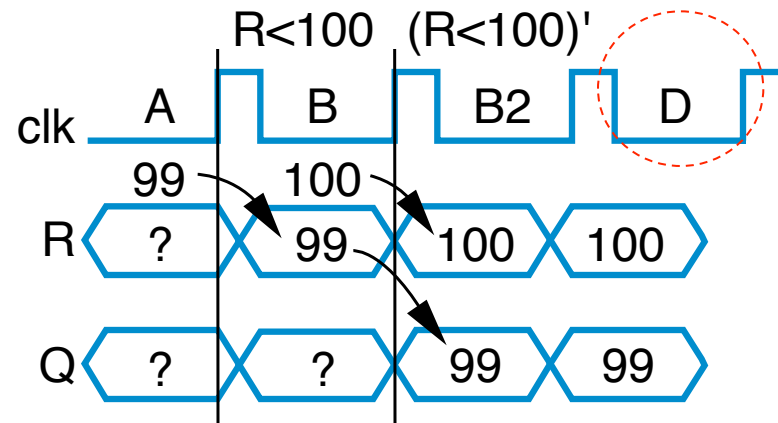
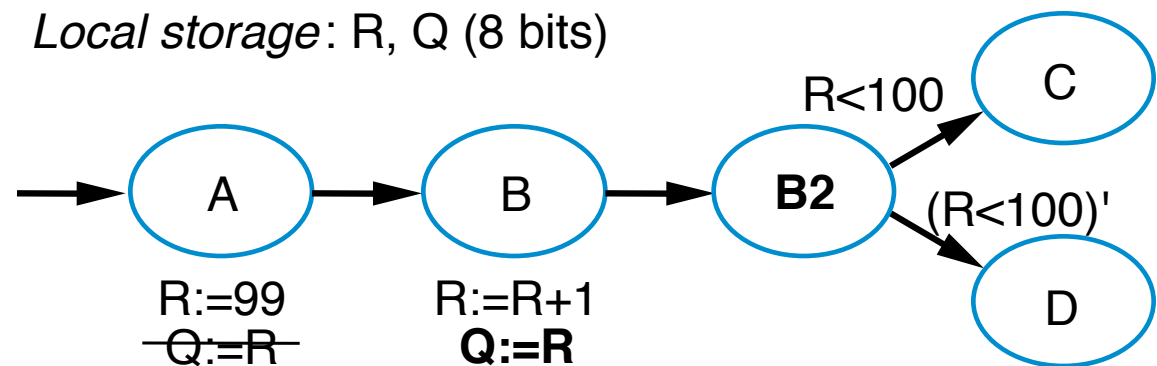
Common RTL Design Pitfall Involving Storage Updates

- Questions
 - Value of Q after state A?
 - Final state is C or D?
- Answers
 - Q is NOT 99 after state A
 - Q is 99 in state B, so final state is C
 - Storage update actions in state occur *simultaneously* on *next* clock edge
 - Thus, order actions are written is irrelevant
 - A's actions same if:
 - $Q:=R$ $R:=99$ or
 - $R:=99$ $Q:=R$



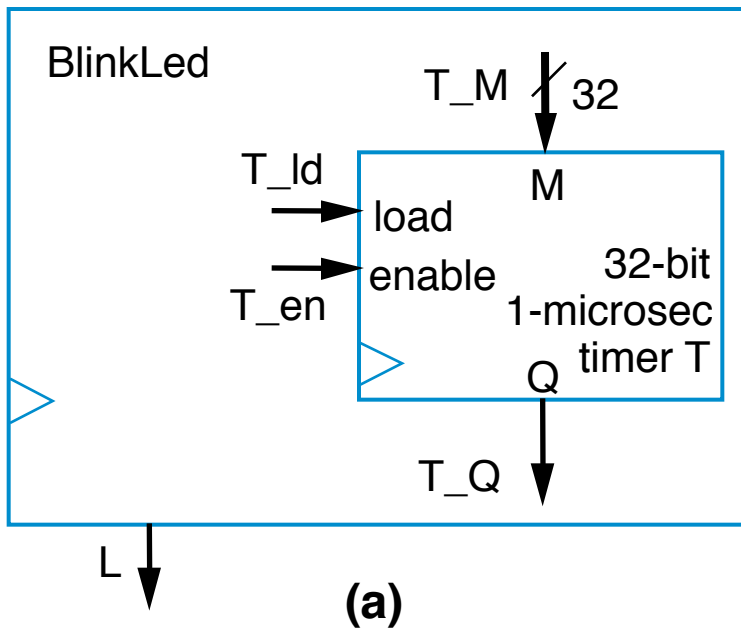
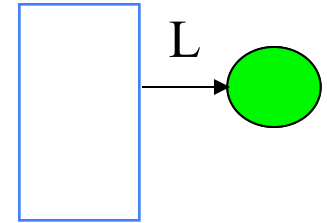
Common RTL Design Pitfall Involving Storage Updates

- New HLSM using extra state so read of R occurs after write of R

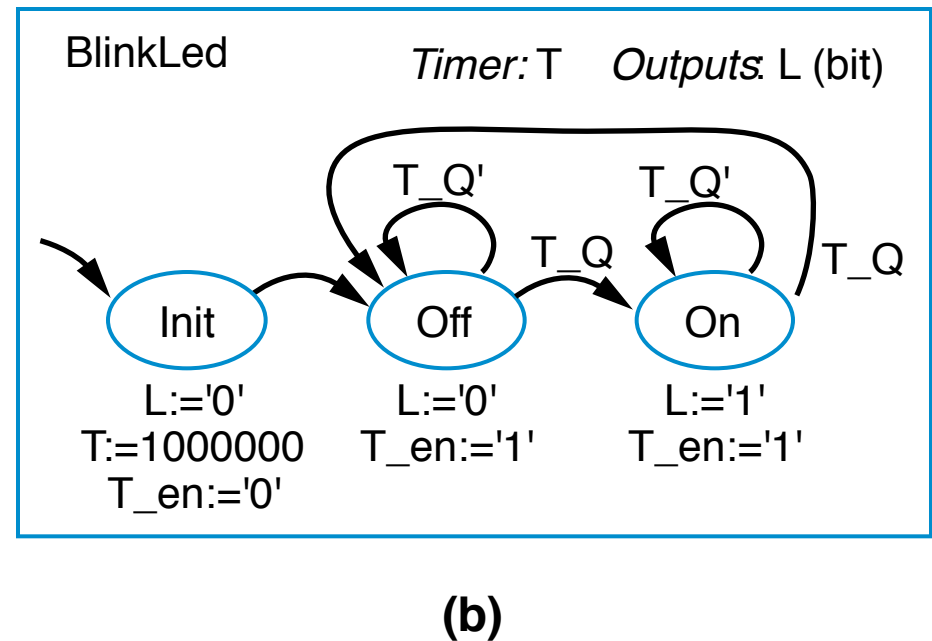


RTL Design Involving a Timer

- Commonly need explicit time intervals
 - Ex: Repeatedly blink LED on 1 second, off 1 second
- Pre-instantiate timer that HLSM can then use



(a) Pre-instantiated timer

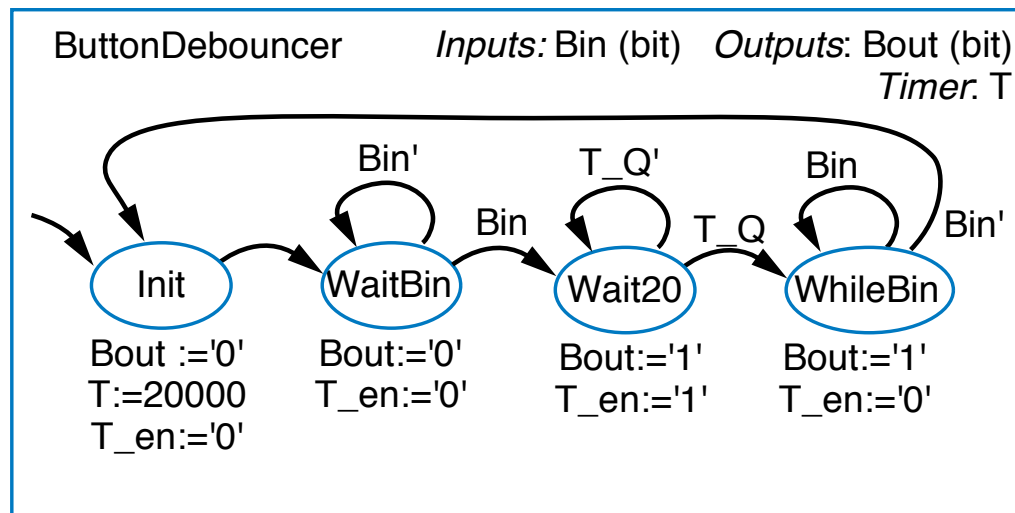
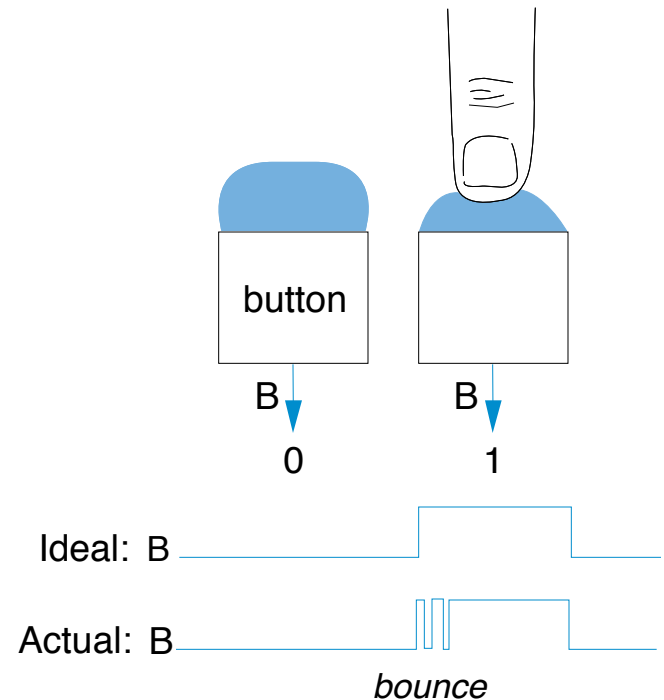


(b) HLSM making use of timer



Button Debouncing

- Press button
 - Ideally, output changes to 1
 - Actually, output bounces
 - Due to mechanical reasons
 - Like ball bouncing when dropped to floor
- Digital circuit can convert actual signal closer to ideal signal



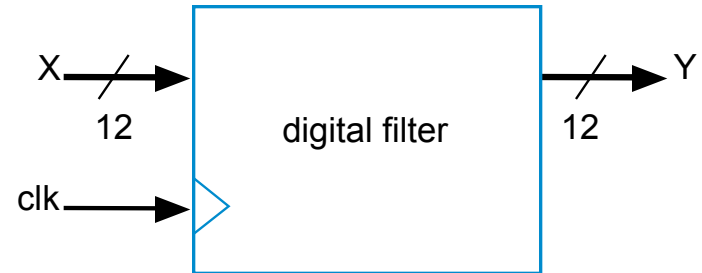
Data Dominated RTL Design Example

- Data dominated design: Extensive DP, simple controller
- Control dominated design: Complex controller, simple DP
- Example: Filter
 - Converts digital input stream to new digital output stream
 - Ex: Remove noise
 - 180, 180, 181, 180, 240, 180, 181
 - 240 is probably noise, filter might replace by 181
 - Simple filter: Output average of last N values
 - Small N : less filtering
 - Large N : more filtering, but less sharp output



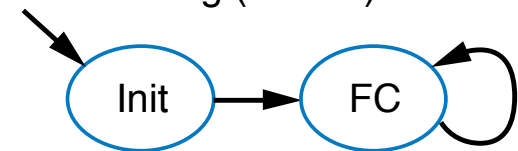
Data Dominated RTL Design Example: FIR Filter

- FIR filter
 - “Finite Impulse Response”
 - Simply a configurable weighted sum of past input values
 - $y(t) = c0*x(t) + c1*x(t-1) + c2*x(t-2)$
 - Above known as “3 tap”
 - Tens of taps more common
 - Very general filter – User sets the constants ($c0$, $c1$, $c2$) to define specific filter
- RTL design
 - Step 1: Create HLSM
 - Very simple states/transitions



$$y(t) = c0*x(t) + c1*x(t-1) + c2*x(t-2)$$

Inputs: X (12 bits) Outputs: Y (12 bits)
Local storage: xt0, xt1, xt2, c0, c1, c2 (12 bits);
Yreg (12 bits)



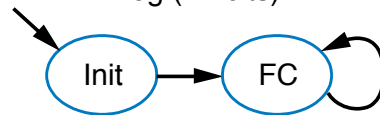
	Yreg := 0	Yreg :=
	xt0 := 0	c0*xt0 +
	xt1 := 0	c1*xt1 +
	xt2 := 0	c2*xt2
	c0 := 3	xt0 := X
	c1 := 2	xt1 := xt0
FIR filter	c2 := 2	xt2 := xt1

Assume constants set to 3, 2, and 2



FIR Filter

Inputs: X (12 bits) Outputs: Y (12 bits)
Local storage: xt0, xt1, xt2, c0, c1, c2 (12 bits);
Yreg (12 bits)



Yreg := 0

xt0 := 0

xt1 := 0

xt2 := 0

c0 := 3

c1 := 2

c2 := 2

Yreg :=

$c0 \cdot xt0 +$

$c1 \cdot xt1 +$

$c2 \cdot xt2$

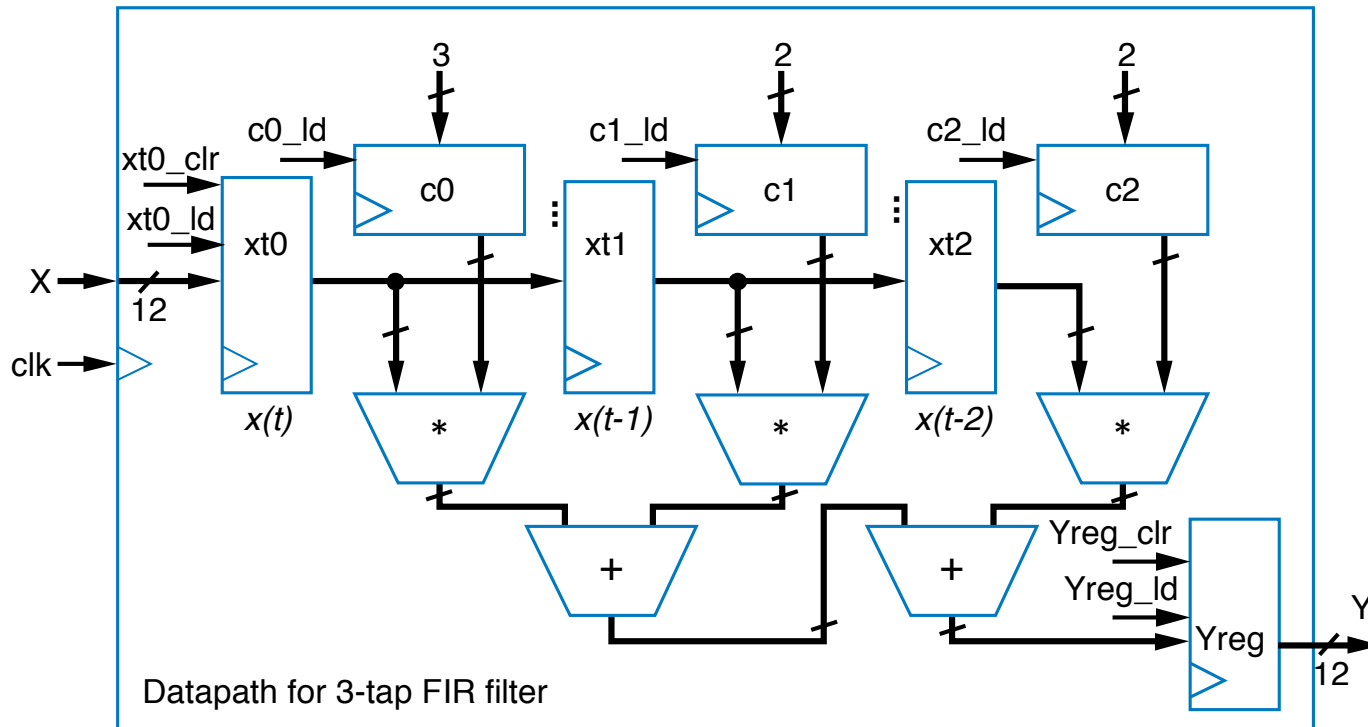
xt0 := X

xt1 := xt0

xt2 := xt1

FIR filter

- Step 2A: Create datapath
- Step 2B: Connect Ctrlr/DP (as earlier examples)
- Step 2C: Derive FSM
 - Set clr and ld lines appropriately



Circuit vs. Microprocessor

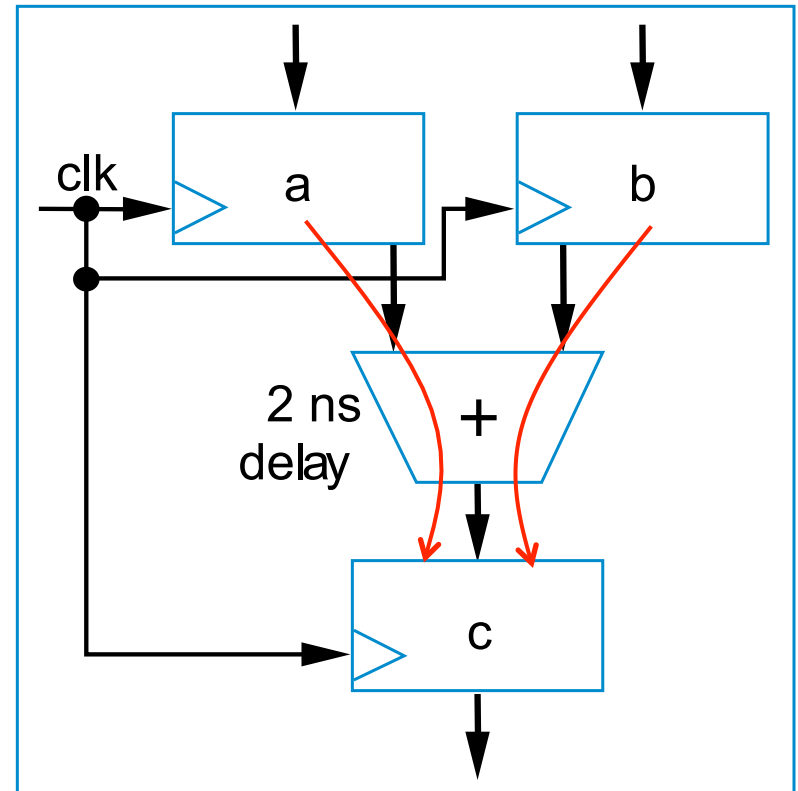
$$y(t) = c_0 \cdot x(t) + c_1 \cdot x(t-1) + c_2 \cdot x(t-2)$$

- Comparing the FIR circuit to microprocessor instructions
 - Microprocessor
 - 100-tap filter: 100 multiplications, 100 additions. Say 2 instructions per multiplication, 2 per addition. Say 10 ns per instruction.
 - $(100 \cdot 2 + 100 \cdot 2) \cdot 10 = 4000$ ns
 - Circuit
 - Assume adder has 2 ns delay, multiplier has 20 ns delay
 - Longest path goes through one multiplier and two adders
 - $20 + 2 + 2 = 24$ ns delay
 - 100-tap filter, following design on previous slide, would have about a 34 ns delay: 1 multiplier and 7 adders on longest path
 - Circuit is more than 100 times faster ($4000/34$). Wow.



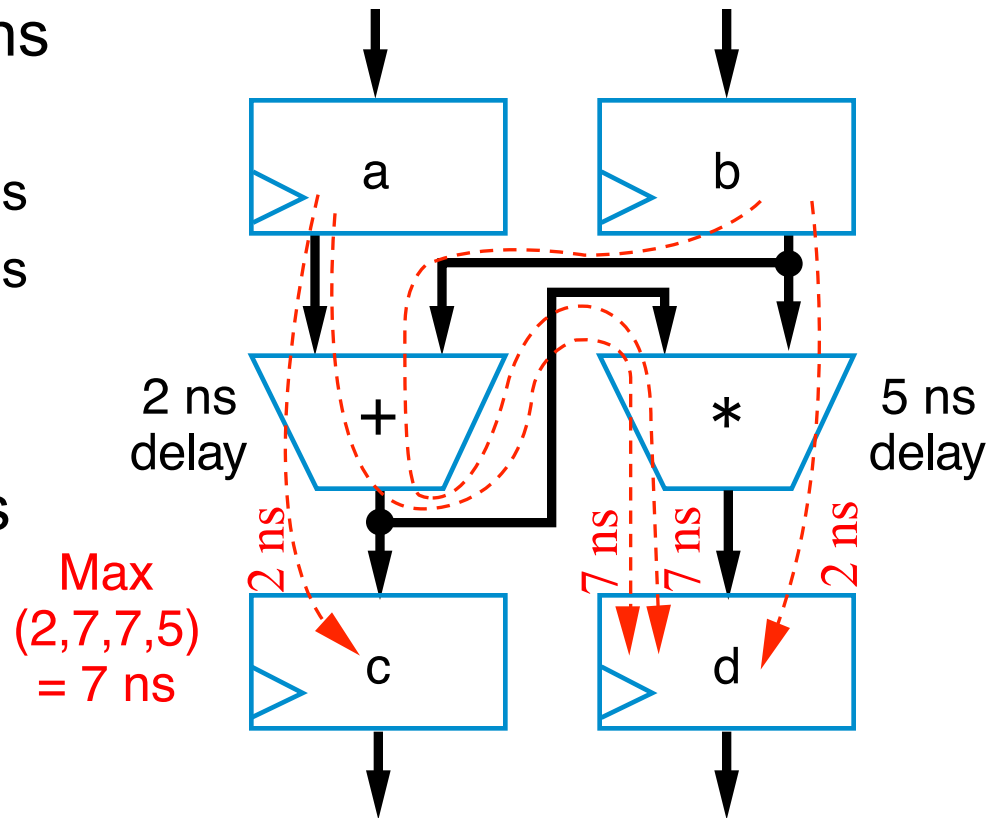
Determining Clock Frequency

- Designers of digital circuits often want fastest performance
 - Means want high clock frequency
- Frequency limited by **longest register-to-register delay**
 - Known as **critical path**
 - If clock is any faster, incorrect data may be stored into register
 - Longest path on right is 2 ns
 - Ignoring wire delays, and register setup and hold times, for simplicity



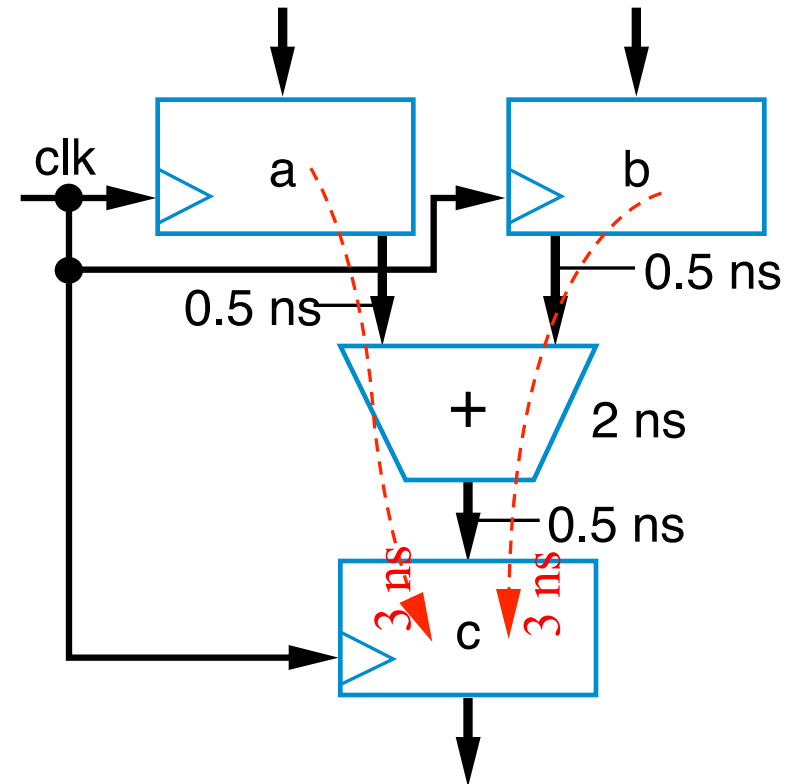
Critical Path

- Example shows four paths
 - a to c through +: 2 ns
 - a to d through + and *: 7 ns
 - b to d through + and *: 7 ns
 - b to d through *: 5 ns
 - b to c through +: 2 ns
- Longest path is thus 7 ns
- Fastest frequency
 - $1 / 7 \text{ ns} = 142 \text{ MHz}$



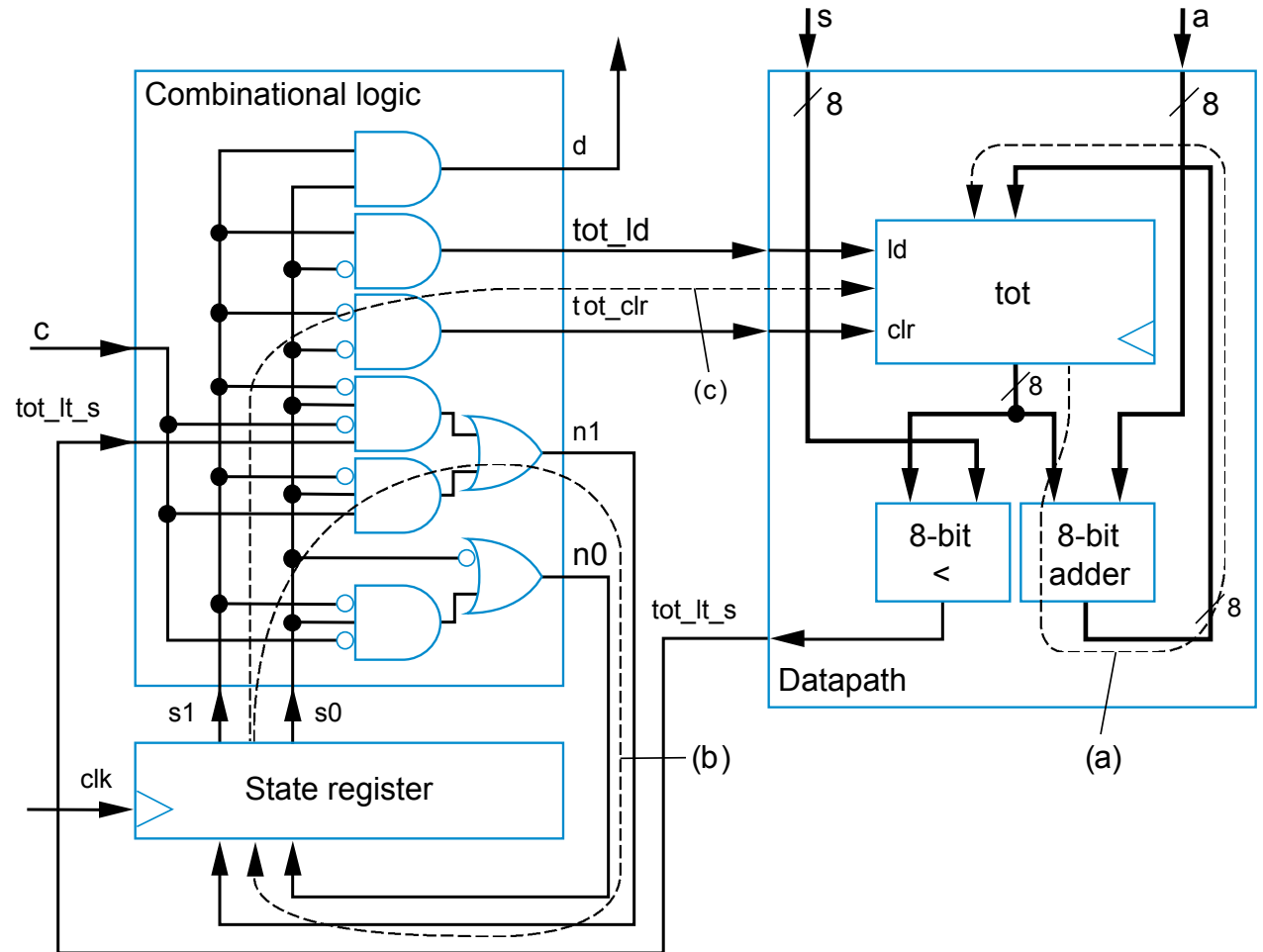
Critical Path Considering Wire Delays

- Real wires have delay too
 - Must include in critical path
- Example shows two paths
 - Each is $0.5 + 2 + 0.5 = 3$ ns
- Trend
 - 1980s/1990s: Wire delays were tiny compared to logic delays
 - But wire delays not shrinking as fast as logic delays
 - Wire delays may even be greater than logic delays!
- Must also consider register setup and hold times, also add to path
- Then add some time to the computed path, just to be safe
 - e.g., if path is 3 ns, say 4 ns instead



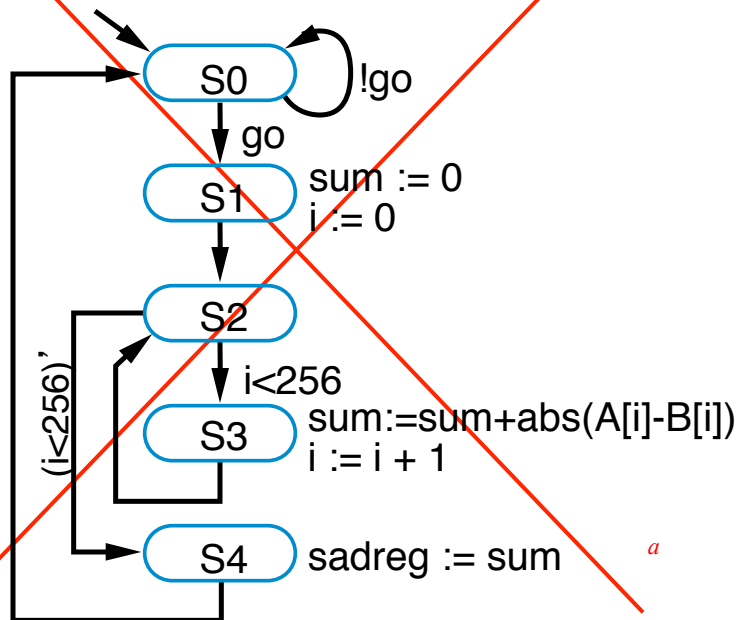
A Circuit May Have Numerous Paths

- Paths can exist
 - In the datapath
 - In the controller
 - Between the controller and datapath
 - May be hundreds or thousands of paths
- Timing analysis tools that evaluate all possible paths automatically very helpful



Behavioral Level Design: C to Gates

Inputs: A, B [256](8 bits); go (bit)
 Outputs: sad (32 bits)
 Local storage: sum, sadreg (32 bits); i (9 bits)



C code

```

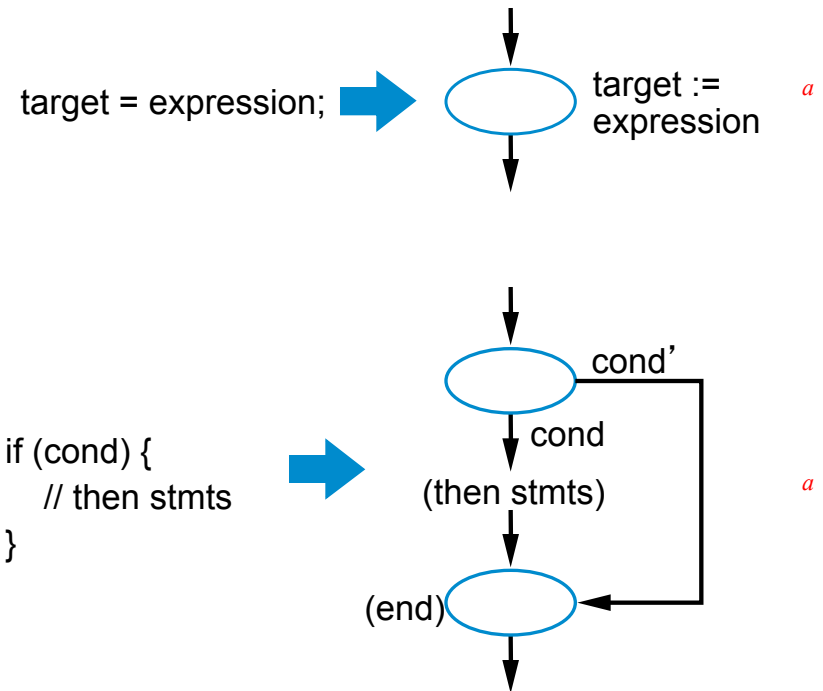
int SAD (byte A[256], byte B[256]) // not quite C syntax
{
    uint sum; short uint I;
    sum = 0;
    i = 0;
    while (i < 256) {
        sum = sum + abs(A[i] - B[i]);
        i = i + 1;
    }
    return sum;
}
  
```

- Earlier sum-of-absolute-differences example
 - Started with high-level state machine
 - C code is an even better starting point -- easier to understand



Converting from C to High-Level State Machine

- Convert each C construct to equivalent states and transitions
- *Assignment* statement
 - Becomes one state with assignment
- *If-then* statement
 - Becomes state with condition check, transitioning to “then” statements if condition true, otherwise to ending state
 - “then” statements would also be converted to states

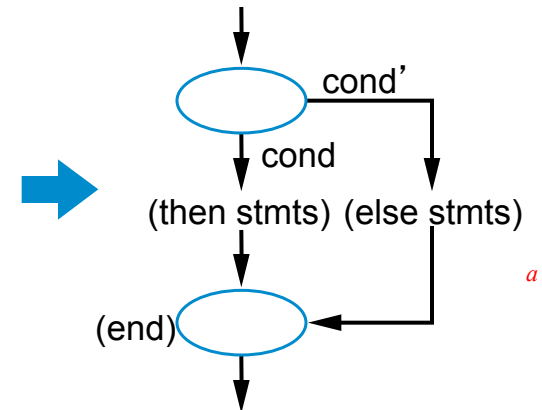


Converting from C to High-Level State Machine

- *If-then-else*

- Becomes state with condition check, transitioning to “then” statements if condition true, or to “else” statements if condition false

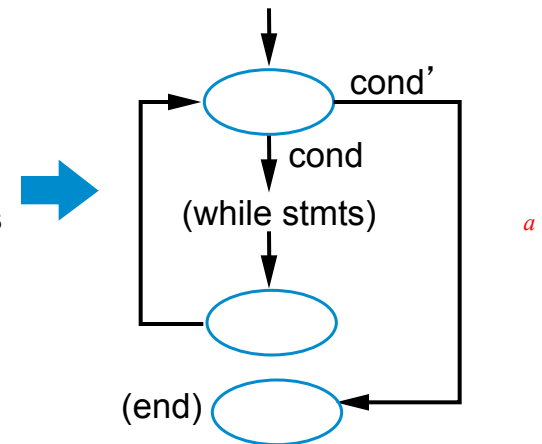
```
if (cond) {  
    // then stmts  
}  
else {  
    // else stmts  
}
```



- *While loop* statement

- Becomes state with condition check, transitioning to while loop's statements if true, then transitioning back to condition check

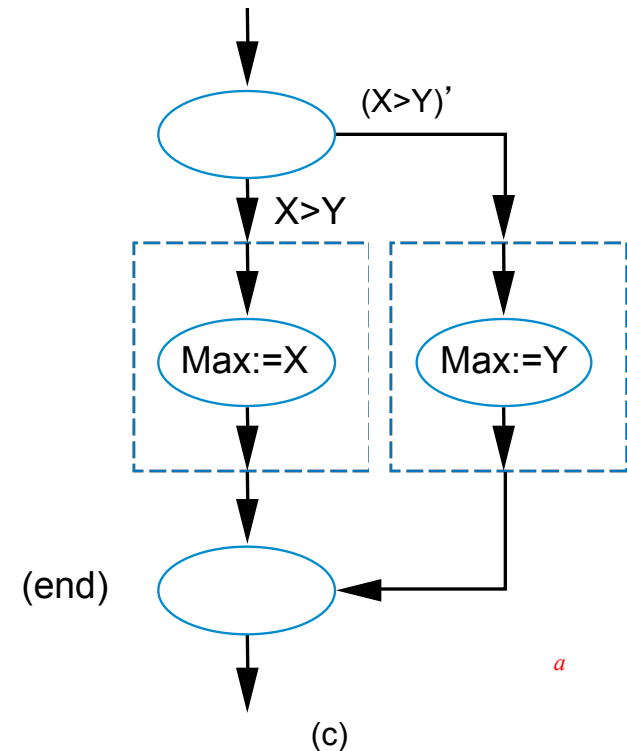
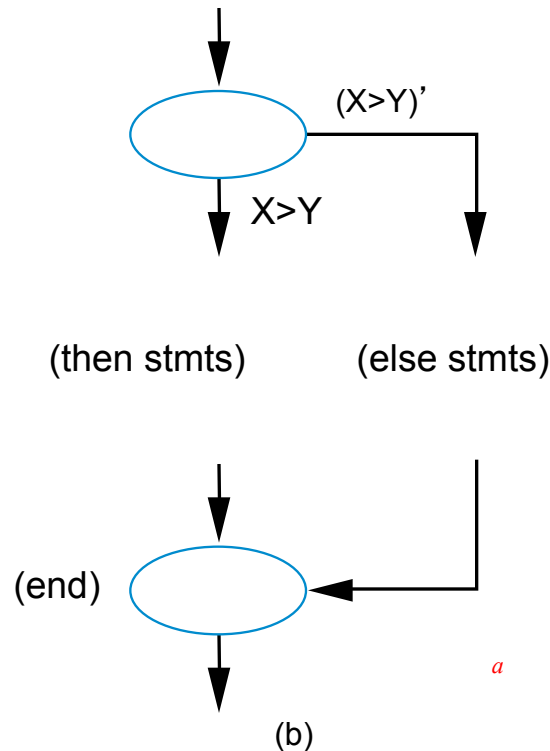
```
while (cond) {  
    // while stmts  
}
```



Simple Example of Converting from C to High-Level State Machine

Inputs: uint X, Y
Outputs: uint Max

```
if (X > Y) {  
    Max = X;  
}  
else {  
    Max = Y;  
}
```



- Simple example: Computing the maximum of two numbers
 - Convert if-then-else statement to states (b)
 - Then convert assignment statements to states (c)



Example: SAD C code to HLSM

Inputs: byte A[256], B[256]
 bit go;
 Output: int sad
 main()
 {

uint sum; short uint i;
 while (1) {

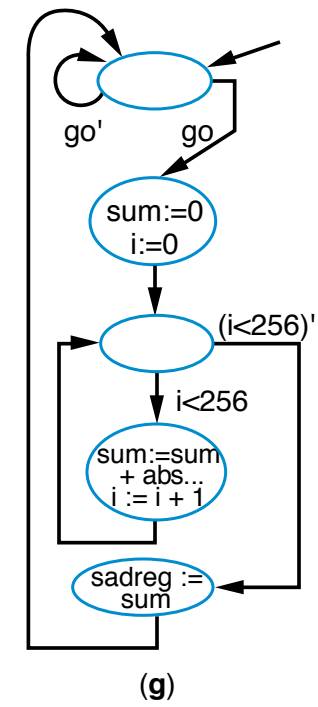
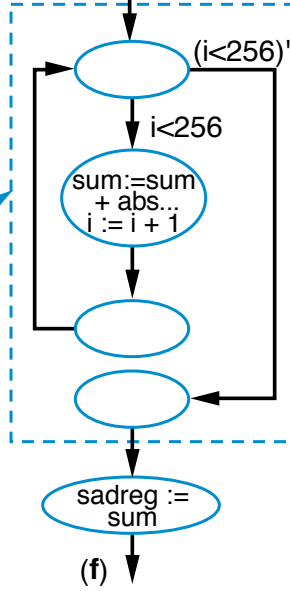
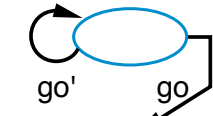
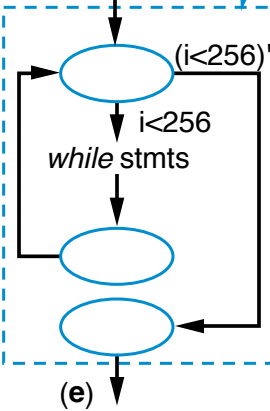
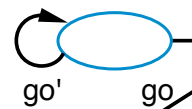
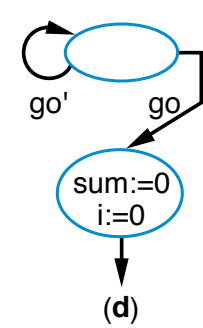
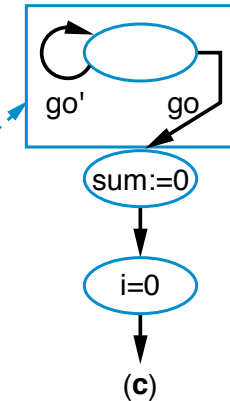
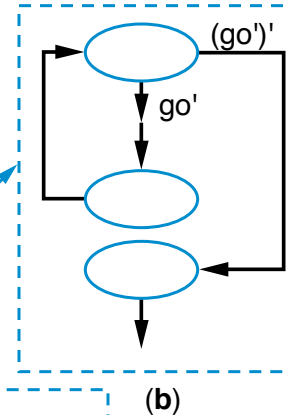
while (!go);
 sum = 0;
 i = 0;

while (i < 256) {
 sum = sum + abs(A[i] - B[i]);
 i = i + 1;

} sad = sum;

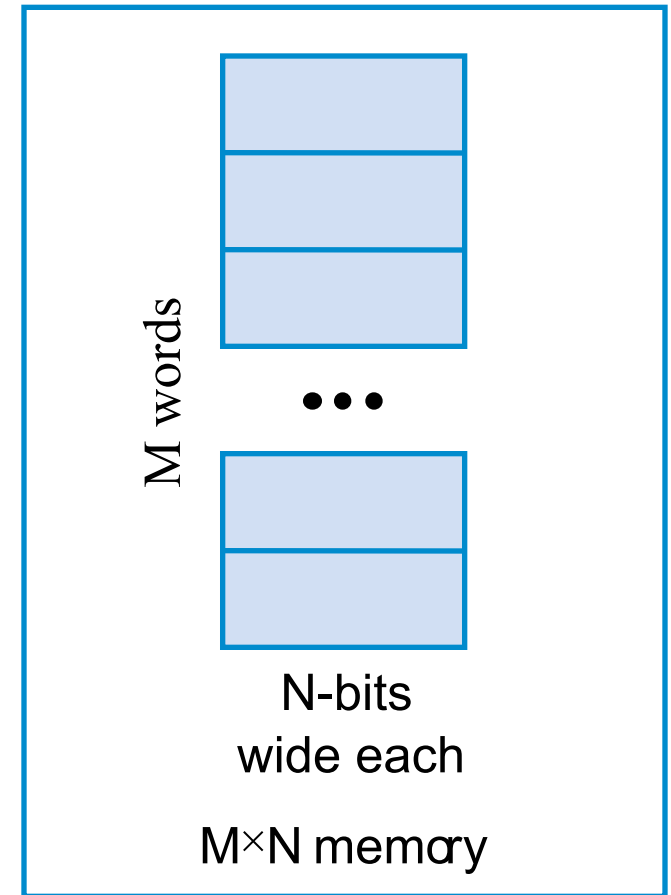
}

(a)



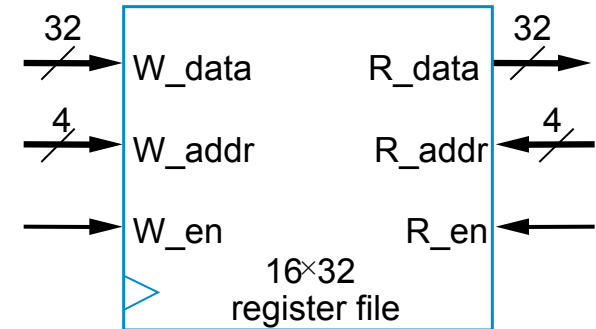
Memory Components

- RTL design instantiates datapath components to create datapath, controlled by a controller
 - Some components are used outside the controller and DP
- ***MxN memory***
 - M words, N bits wide each
- Several varieties of memory, which we now introduce

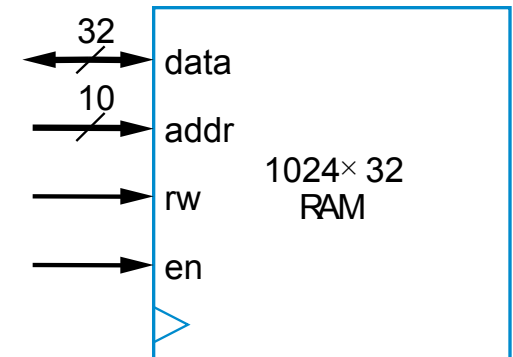


Random Access Memory (RAM)

- RAM – Readable and writable memory
 - “Random access memory”
 - Strange name—Created several decades ago to contrast with sequentially-accessed storage like tape drives
 - Logically same as register file—Memory with address inputs, data inputs/outputs, and control
 - RAM usually one port; RF usually two or more
 - RAM vs. RF
 - RAM typically larger than *about* 512 or 1024 words
 - RAM typically stores bits using a bit storage approach that is more efficient than a flip-flop
 - RAM typically implemented on a chip in a square rather than rectangular shape—keeps longest wires (hence delay) short



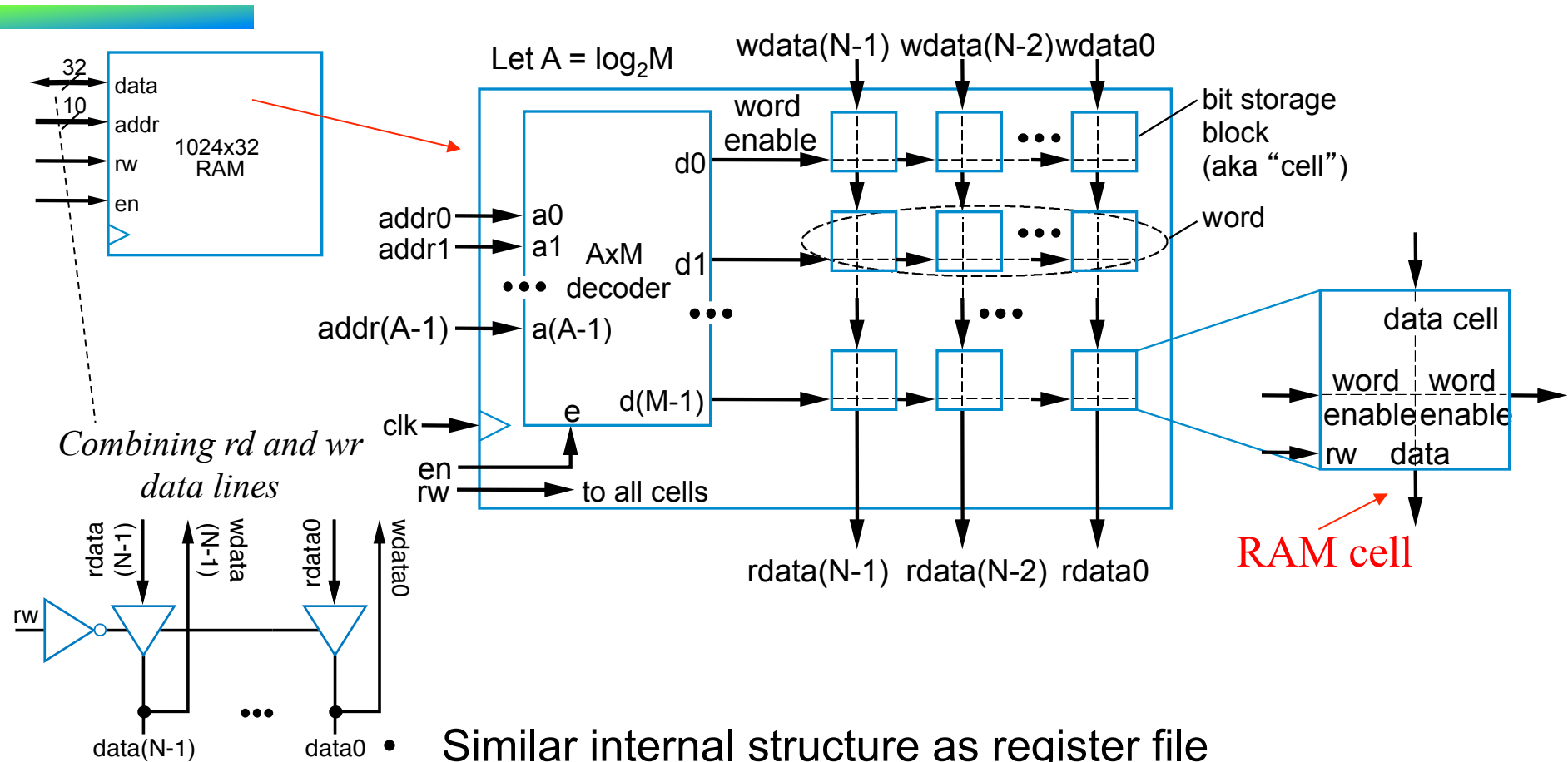
Register file from Chpt. 4



RAM block symbol



RAM Internal Structure

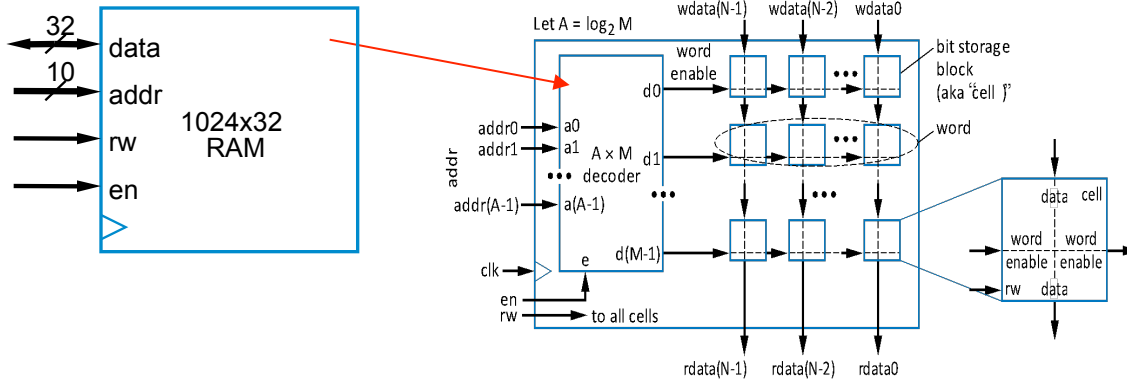


Similar internal structure as register file

- Decoder enables appropriate word based on address inputs
- *rw* controls whether cell is written or read
- *rd* and *wr* data lines typically combined
- Let's see what's inside each RAM cell

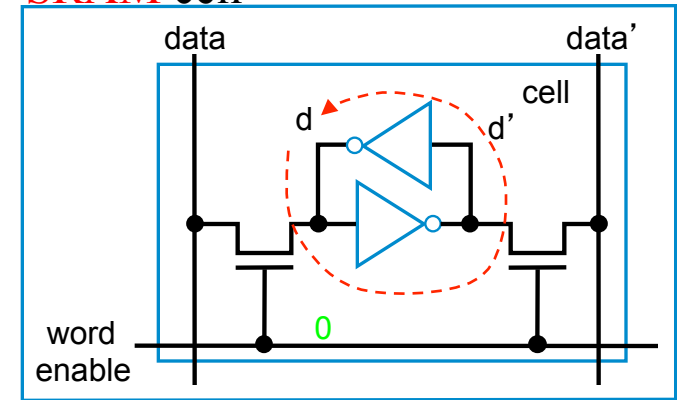


Static RAM (SRAM)

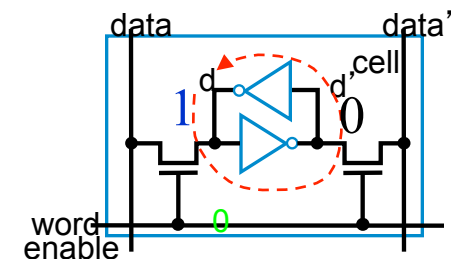
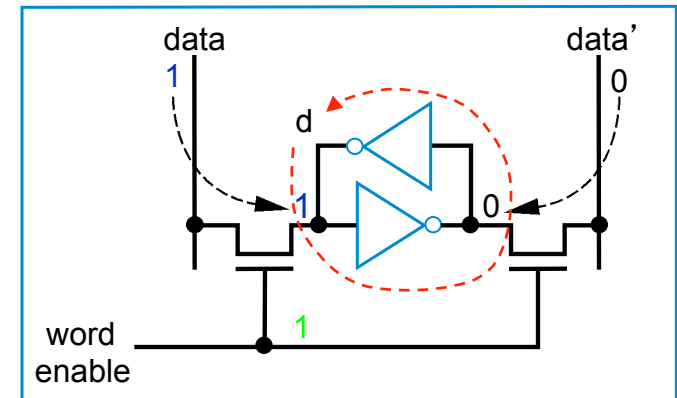


- “Static” RAM cell
 - 6 transistors (recall inverter is 2 transistors)
 - Writing this cell
 - *word enable* input comes from decoder
 - When 0, value *d* loops around inverters
 - That loop is where a bit stays stored
 - When 1, the *data* bit value enters the loop
 - *data* is the bit to be stored in this cell
 - *data'* enters on other side
 - Example shows a “1” being written into cell

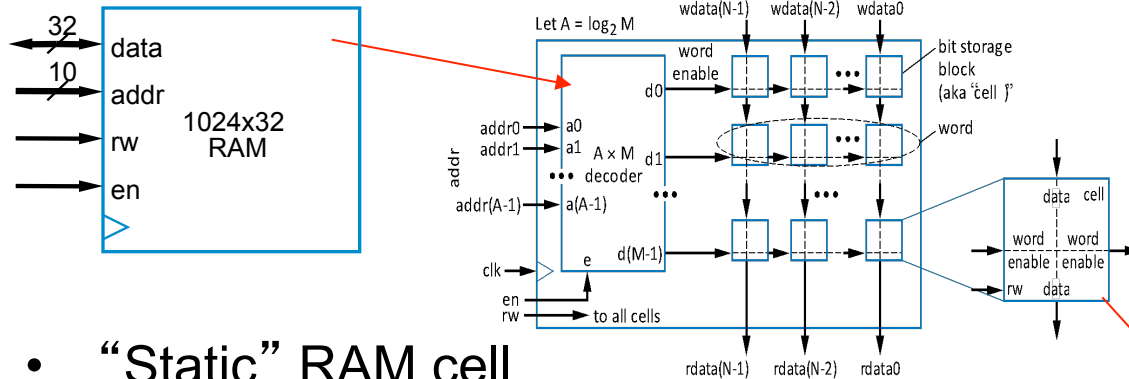
SRAM cell



SRAM cell

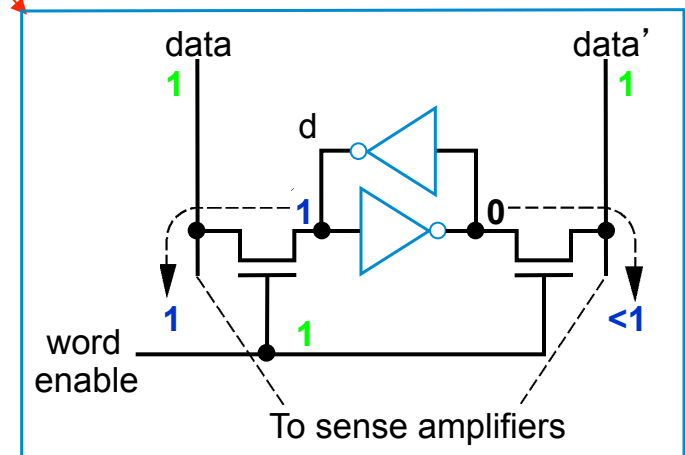


Static RAM (SRAM)

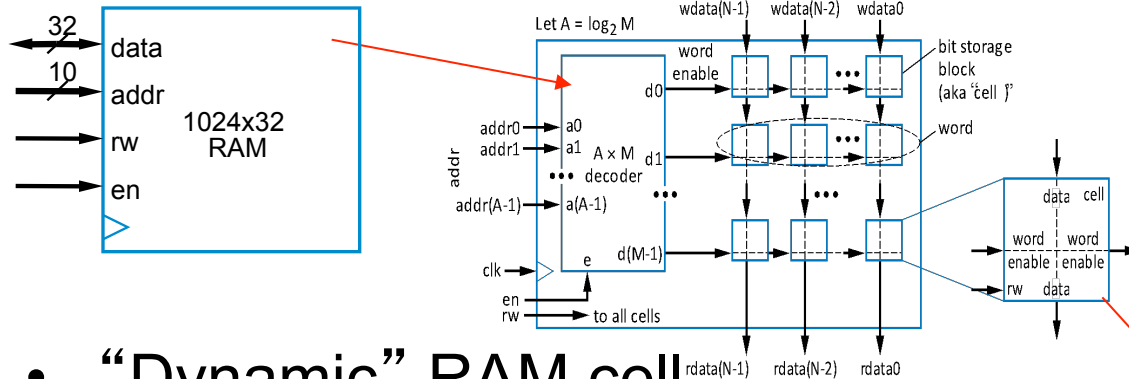


- “Static” RAM cell
 - Reading this cell
 - Somewhat trickier
 - When rw set to read, the RAM logic sets both *data* and *data'* to 1
 - The stored bit *d* will pull either the left line or the right bit down slightly below 1
 - “Sense amplifiers” detect which side is slightly pulled down
 - The electrical description of SRAM is really beyond our scope – just general idea here, mainly to contrast with *DRAM*...

SRAM cell

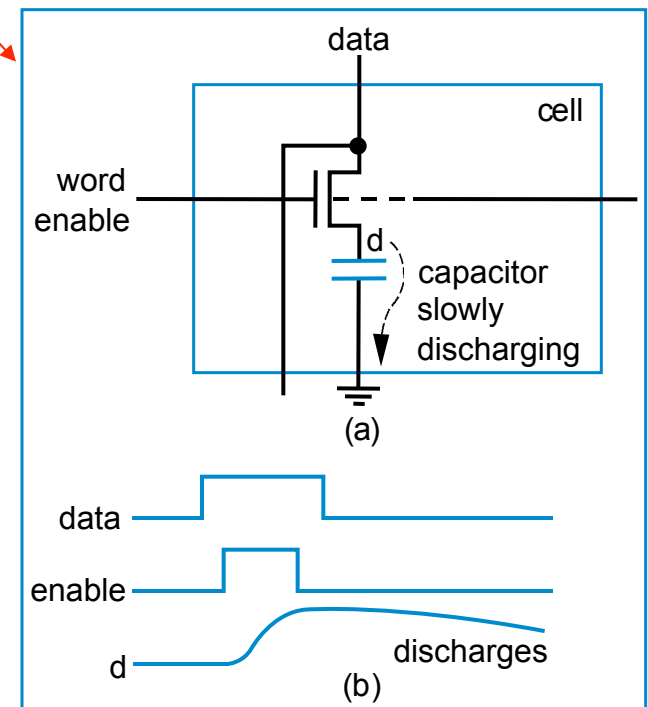


Dynamic RAM (DRAM)



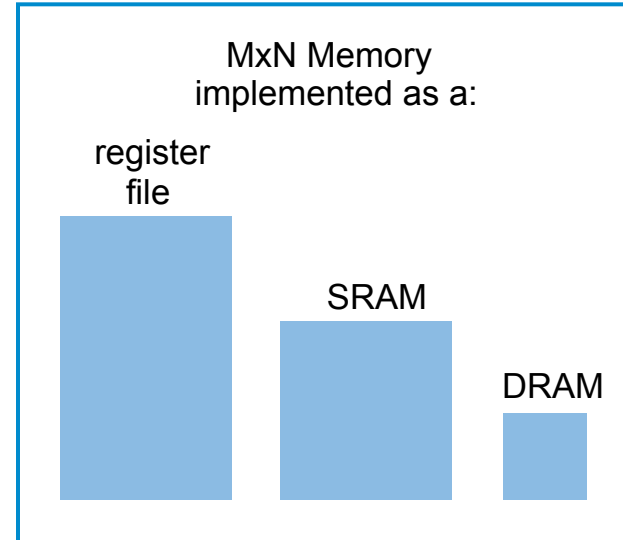
- “Dynamic” RAM cell
 - 1 transistor (rather than 6)
 - Relies on *large* capacitor to store bit
 - Write: Transistor conducts, data voltage level gets stored on top plate of capacitor
 - Read: Just look at value of d
 - Problem: Capacitor discharges over time
 - Must “refresh” regularly, by reading d and then writing it right back

DRAM cell



Comparing Memory Types

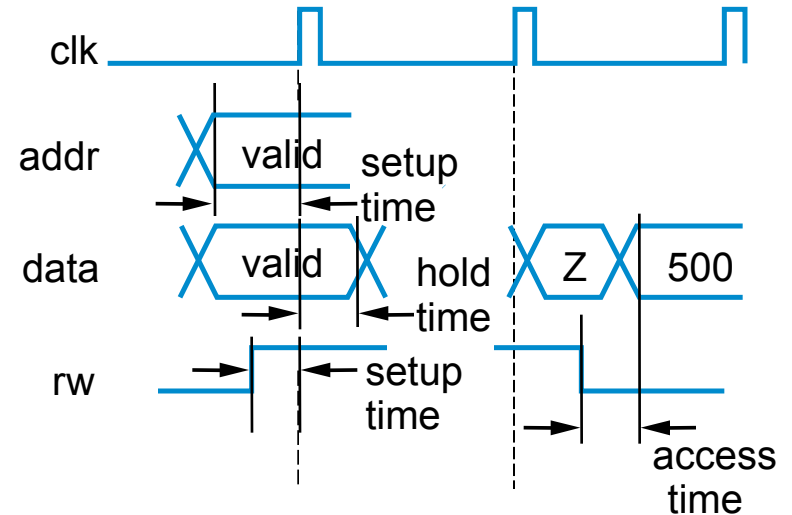
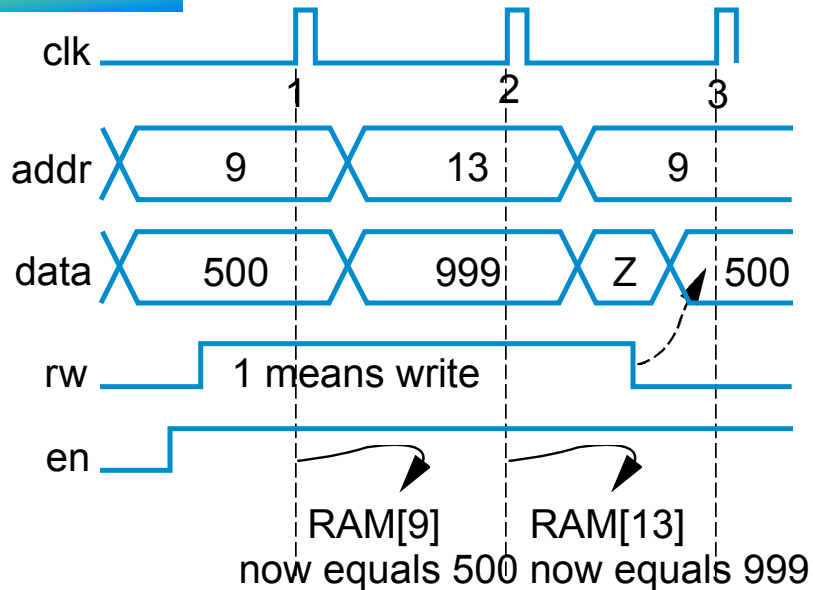
- Register file
 - Fastest
 - But biggest size
- SRAM (caches in PCs)
 - Fast
 - More compact than register file
- DRAM (main RAM memory in PCs)
 - Slowest
 - And refreshing takes time
 - But very compact
- Use register file for small items, SRAM for large items, and DRAM for huge items
 - Note: DRAM's big capacitor requires a special chip design process, so DRAM is often a separate chip



Size comparison for same number of bits (not to scale)



Reading and Writing a RAM

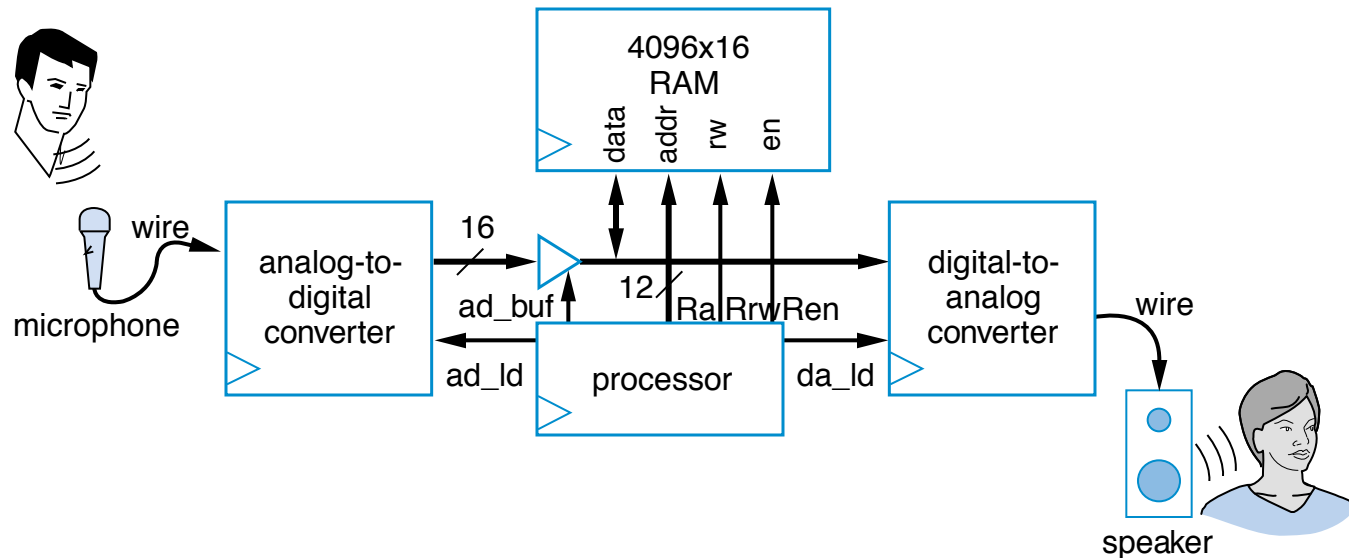


- Writing
 - Put address on *addr* lines, data on *data* lines, set *rw*=1, *en*=1
- Reading
 - Set *addr* and *en* lines, but put nothing (Z) on *data* lines, set *rw*=0
 - Data will appear on *data* lines
- Don't forget to obey setup and hold times
 - In short – keep inputs stable before and after a clock edge

(b)



RAM Example: Digital Sound Recorder

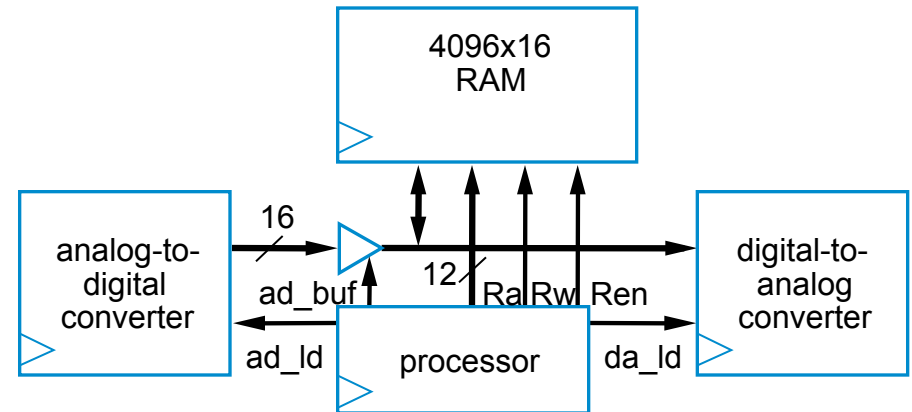


- Behavior
 - Record: Digitize sound, store as series of 4096 12-bit digital values in RAM
 - We'll use a 4096x16 RAM (12-bit wide RAM not common)
 - Play back later
 - Common behavior in telephone answering machine, toys, voice recorders
- To record, processor should read a-to-d, store read values into successive RAM words
 - To play, processor should read successive RAM words and enable d-to-a



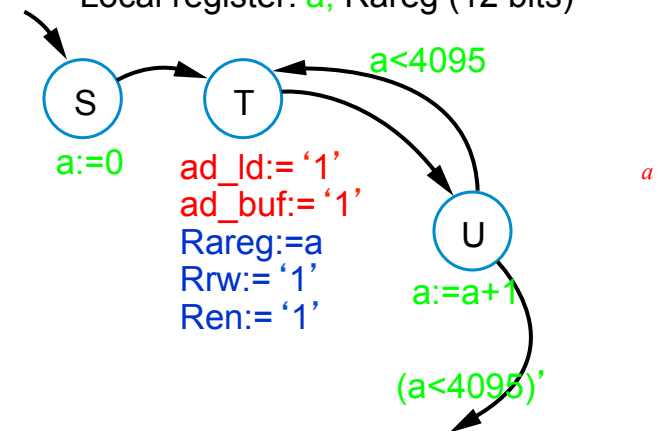
RAM Example: Digital Sound Recorder

- RTL design of processor
 - Create HLSM
 - Begin with the *record* behavior
 - Create local storage *a*
 - Stores current address, ranges from 0 to 4095 (thus need 12 bits)
 - Create state machine that counts from 0 to 4095 using *a*
 - For each *a*
 - Read analog-to-digital conv.
 - » $ad_ld := '1'$,
 $ad_buf := '1'$
 - Write to RAM at address *a*
 - » $Rareg := a$, $Rrw := '1'$,
 $Ren := '1'$



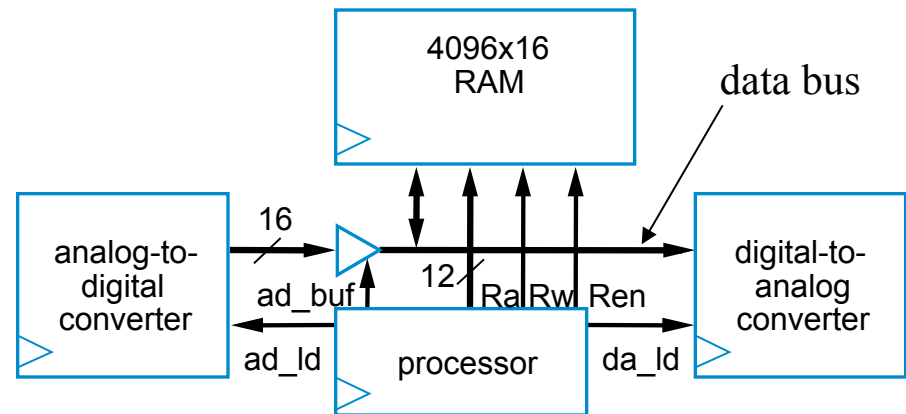
Record behavior

Local register: *a*, Rareg (12 bits)

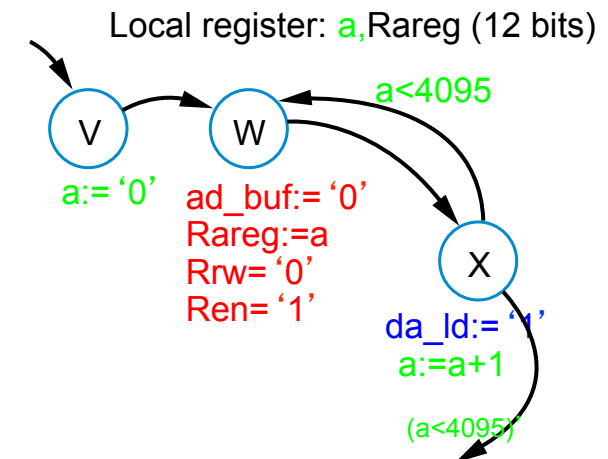


RAM Example: Digital Sound Recorder

- Now create *play* behavior
- Use local register *a* again, create **state machine that counts from 0 to 4095** again
 - For each *a*
 - **Read RAM**
 - **Write to digital-to-analog conv.**
 - Note: Must write d-to-a one cycle *after* reading RAM, when the read data is available on the data bus
- The record and play state machines would be parts of a larger state machine controlled by signals that determine when to record or play

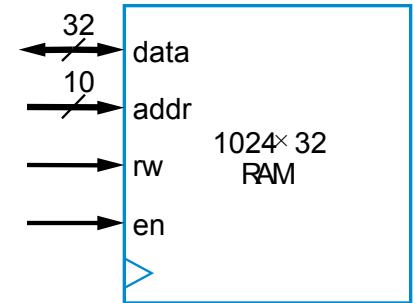


Play behavior

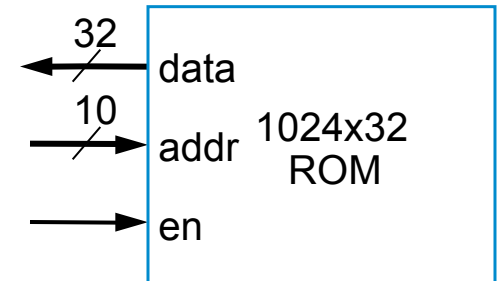


Read-Only Memory – ROM

- Memory that can only be read from, not written to
 - Data lines are output only
 - No need for *rw* input
- Advantages over RAM
 - Compact: May be smaller
 - **Nonvolatile**: Saves bits even if power supply is turned off
 - Speed: May be faster (especially than DRAM)
 - Low power: Doesn't need power supply to save bits, so can extend battery life,
 - Embedded systems, cameras, phones, firmware
- Choose ROM over RAM if stored data won't change (or won't change often)
 - For example, a table of Celsius to Fahrenheit conversions in a digital thermometer



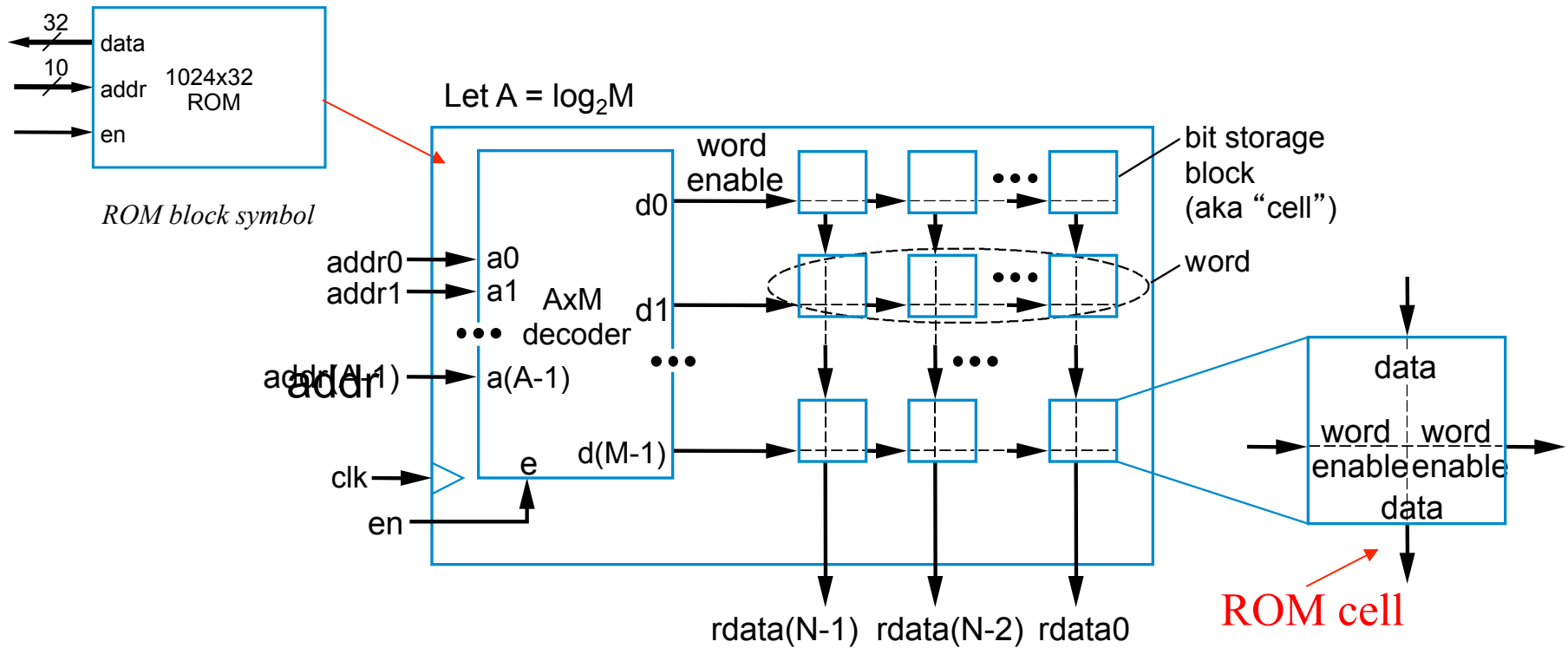
RAM block symbol



ROM block symbol



Read-Only Memory – ROM

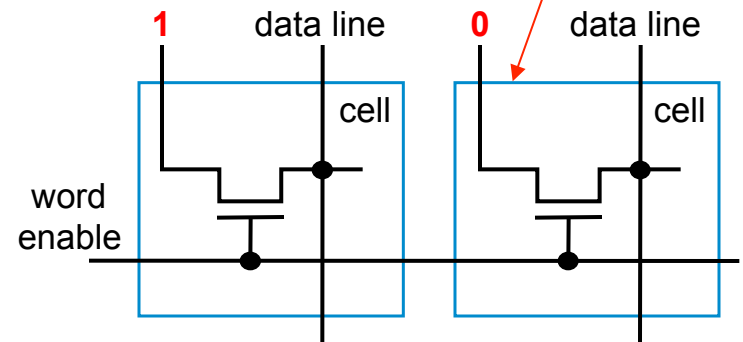
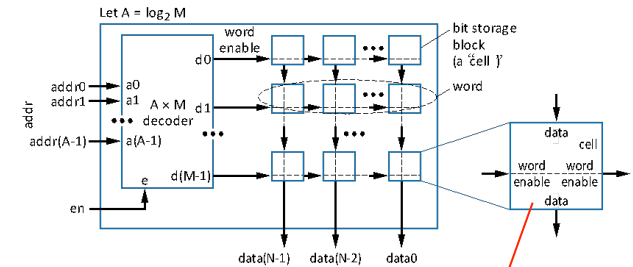


- Internal logical structure similar to RAM, without the data input lines



ROM Types

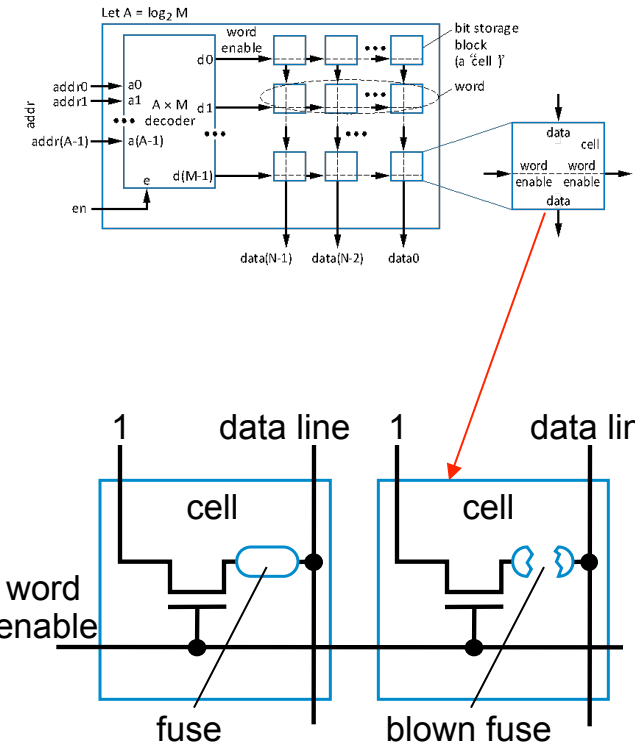
- If a ROM can only be read, how are the stored bits stored in the first place?
 - Storing bits in a ROM known as *programming*
 - Several methods
- **Mask-programmed ROM**
 - Bits are hardwired as 0s or 1s during chip manufacturing
 - 2-bit word on right stores “10”
 - word enable (from decoder) simply passes the hardwired value through transistor
 - Notice how compact, and fast, this memory would be



ROM Types

- **Fuse-Based Programmable ROM**

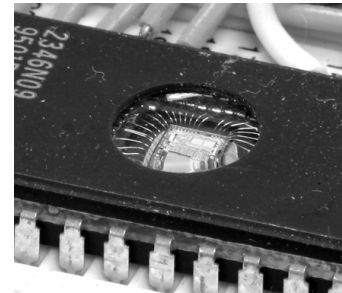
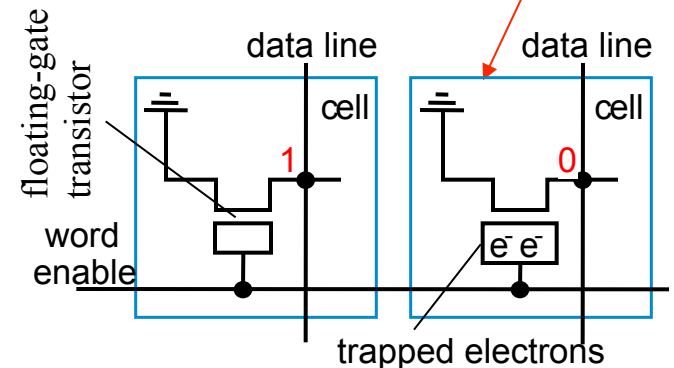
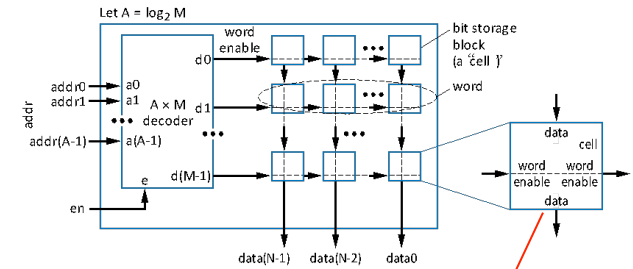
- Each cell has a fuse
- A special device, known as a programmer, blows certain fuses (using higher-than-normal voltage)
 - Those cells will be read as 0s (involving some special electronics)
 - Cells with unblown fuses will be read as 1s
 - 2-bit word on right stores “10”
- Also known as **One-Time Programmable (OTP) ROM**



ROM Types

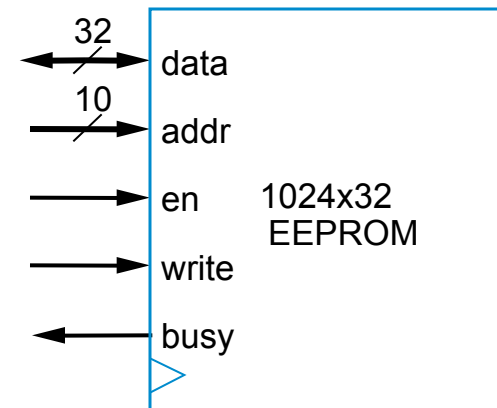
- **Erasable Programmable ROM (EPROM)**

- Uses “floating-gate transistor” in each cell
- Special programmer device uses higher-than-normal voltage to cause electrons to *tunnel* into the gate
 - Electrons become trapped in the gate
 - Only done for cells that should store 0
 - Other cells (without electrons trapped in gate) will be 1
 - 2-bit word on right stores “10”
 - Details beyond our scope – just general idea is necessary here
- To erase, shine ultraviolet light onto chip
 - Gives trapped electrons energy to escape
 - Requires chip package to have window

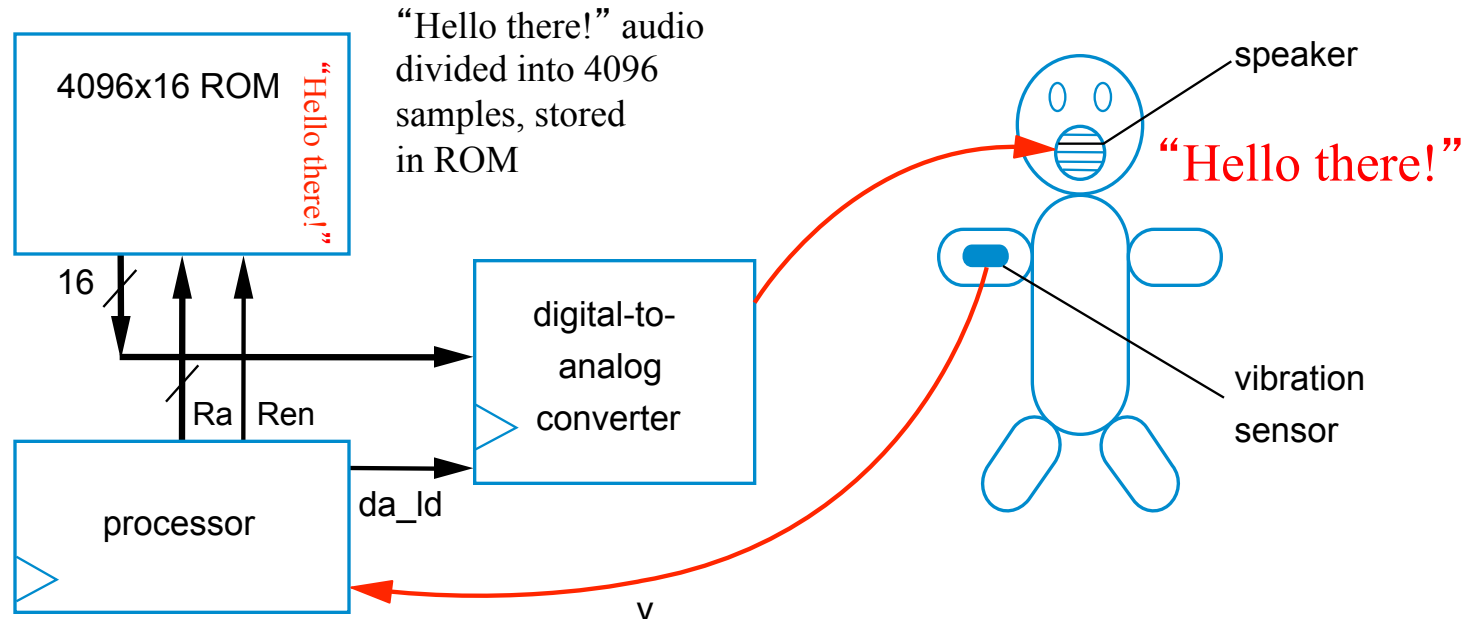


ROM Types

- **Electronically-Erasable Programmable ROM (EEPROM)**
 - Similar to EPROM
 - Uses floating-gate transistor, electronic programming to trap electrons in certain cells
 - But erasing done *electronically*, not using UV light
 - Erasing done one word at a time
- **Flash memory**
 - Like EEPROM, but all words (or large blocks of words) can be erased *simultaneously*
 - Became very common starting in late 1990s
- Both types are *in-system programmable*
 - Can be programmed with new stored bits while in the system in which the ROM operates
 - Requires bi-directional data lines, and write control input
 - Also need busy output to indicate that erasing is in progress – erasing takes some time



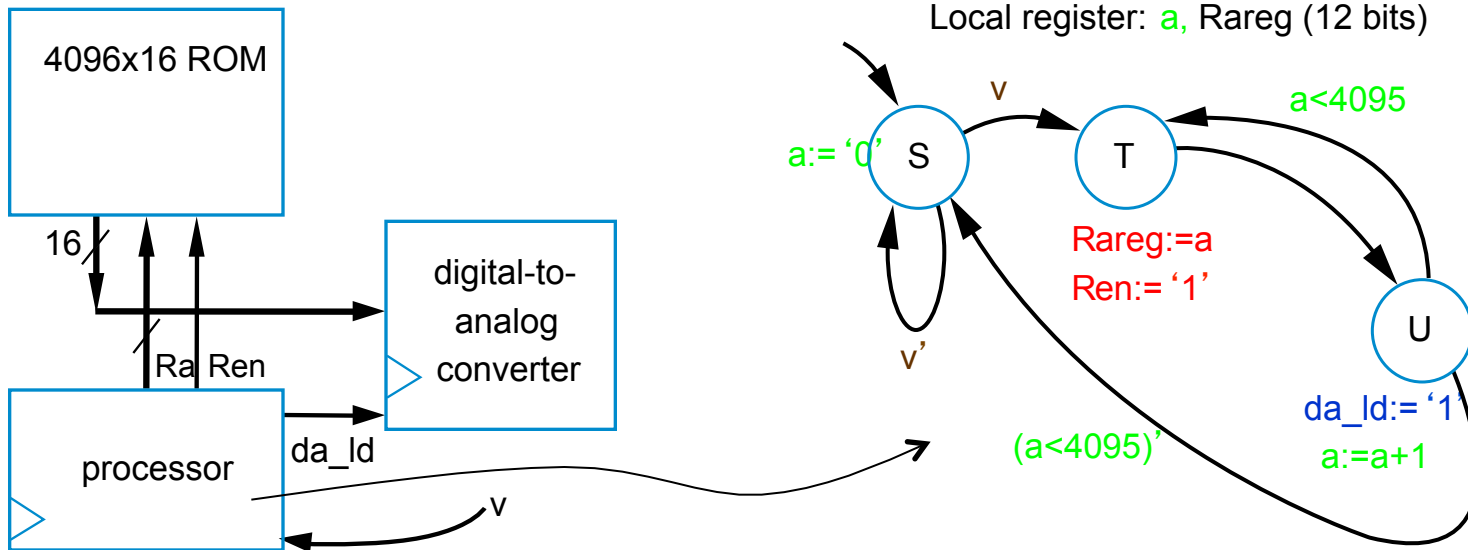
ROM Example: Talking Doll



- Doll plays prerecorded message, triggered by vibration
 - Message must be stored without power supply → Use a ROM, not a RAM, because ROM is nonvolatile
 - And because message will never change, may use a mask-programmed ROM or OTP ROM
 - Processor should wait for vibration ($v=1$), then read words 0 to 4095 from the ROM, writing each to the d-to-a



ROM Example: Talking Doll



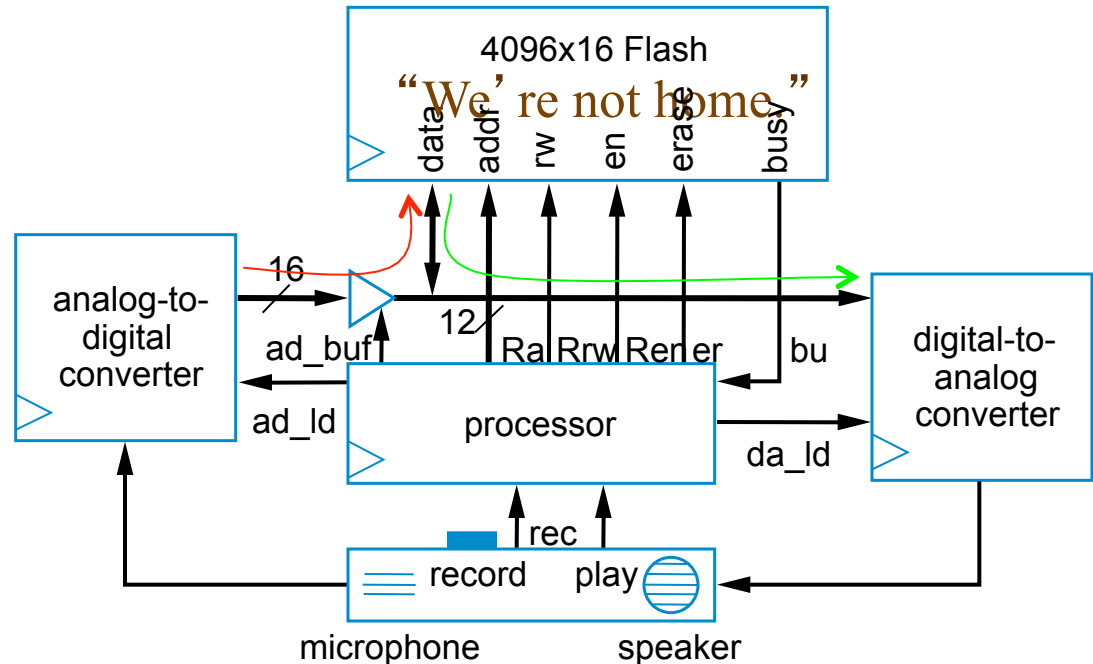
• HLASM

- Create state machine that **waits** for $v=1$, and then **counts** from 0 to 4095 using a local storage a
- For each a , **read ROM**, **write** to digital-to-analog converter



ROM Example: Digital Telephone Answering Machine Using a Flash Memory

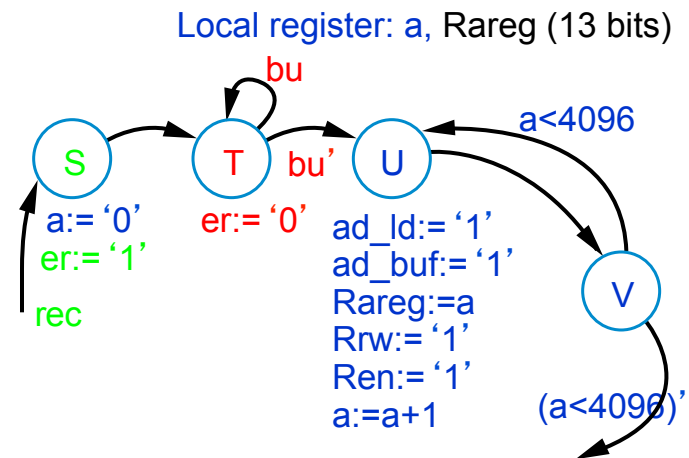
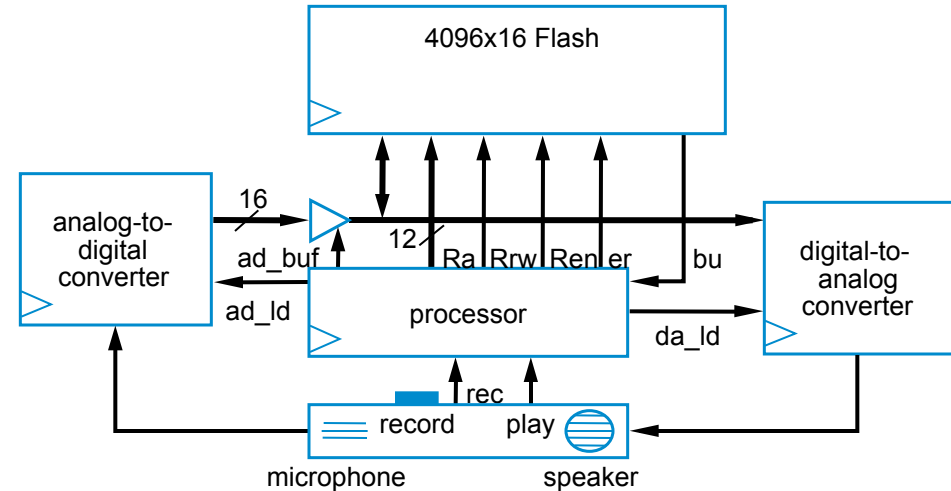
- Want to record the **outgoing announcement**
 - When $rec=1$, **record** digitized sound in locations 0 to 4095
 - When $play=1$, **play** those stored sounds to digital-to-analog converter
- What type of memory?
 - Should store without power supply – ROM, not RAM
 - Should be in-system programmable – EEPROM or Flash, not EPROM, OTP ROM, or mask-programmed ROM
 - Will always erase entire memory when reprogramming – Flash better than EEPROM



ROM Example: Digital Telephone Answering Machine Using a Flash Memory

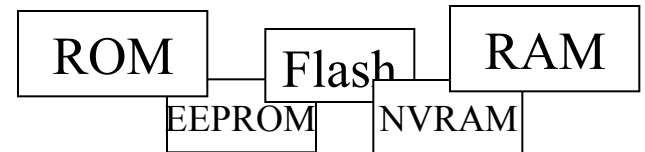
- HLSM

- Once $rec=1$, begin erasing flash by setting $er=1$
- Wait for flash to finish erasing by waiting for $bu=0$
- Execute loop that sets local register a from 0 to 4095, reading analog-to-digital converter and writing to flash for each a



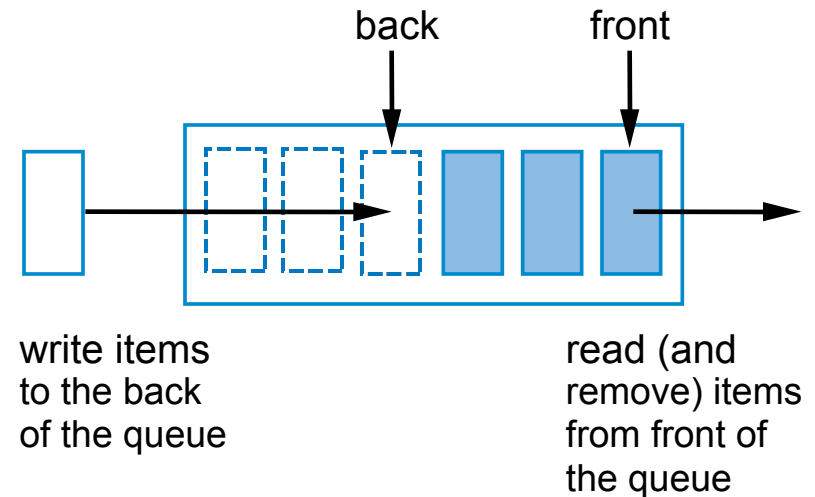
Blurring of Distinction Between ROM and RAM

- We said that
 - RAM is readable and writable
 - ROM is read-only
- But some ROMs act almost like RAMs
 - EEPROM and Flash are in-system programmable
 - Essentially means that writes are slow
 - Also, number of writes may be limited (perhaps a few million times)
- And, some RAMs act almost like ROMs
 - Non-volatile RAMs: Can save their data without the power supply
 - One type: Built-in battery, may work for up to 10 years
 - Another type: Includes ROM backup for RAM – controller writes RAM contents to ROM before turning off
- New memory technologies evolving that merge RAM and ROM benefits
 - e.g., MRAM
- Bottom line
 - Lot of choices available to designer, must find best fit with design goals



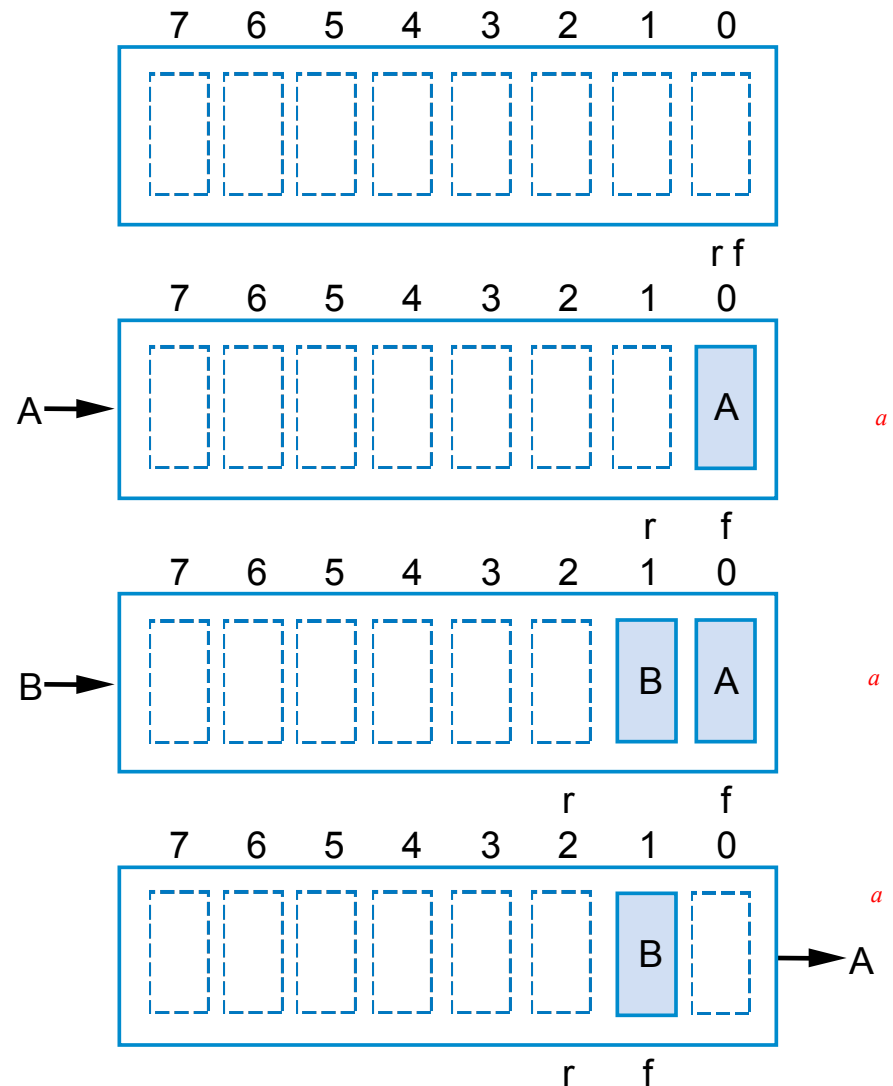
Queues (FIFOs)

- A queue is another component sometimes used during RTL design
- **Queue**: A list written to at the back, read from the front
 - Like a list of waiting restaurant customers
- Writing called a **push**, reading called a **pop**
- Because first item written into a queue will be the first item read out, also called a **FIFO** (first-in-first-out)



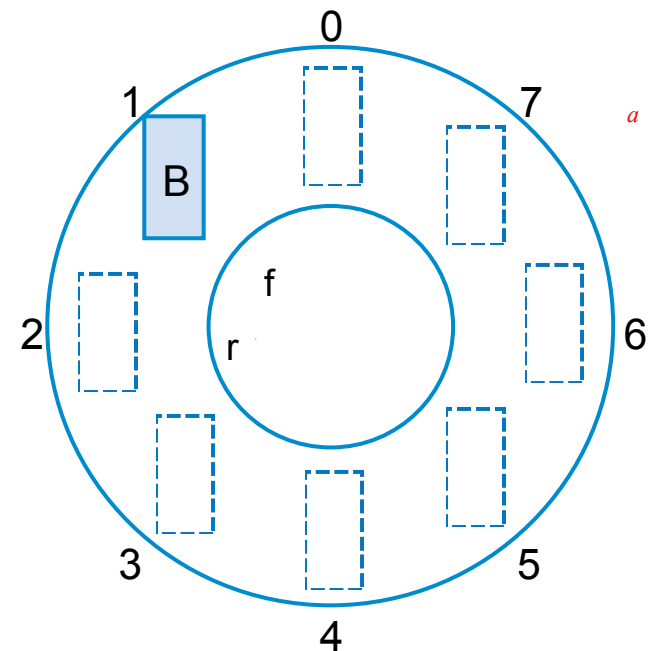
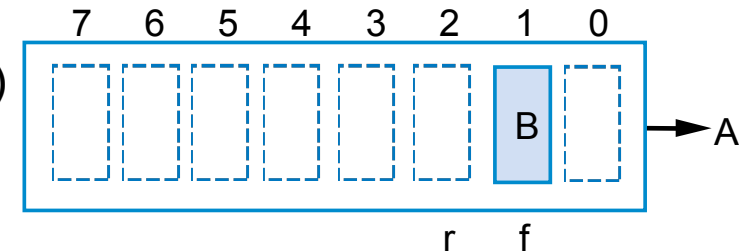
Queues

- Queue has addresses, and two pointers: *rear* and *front*
 - Initially both point to 0
- Push (write)
 - Item written to address pointed to by *rear*
 - rear* incremented
- Pop (read)
 - Item read from address pointed to by *front*
 - front* incremented
- If front or rear reaches 7, next (incremented) value should be 0 (for a queue with addresses 0 to 7)



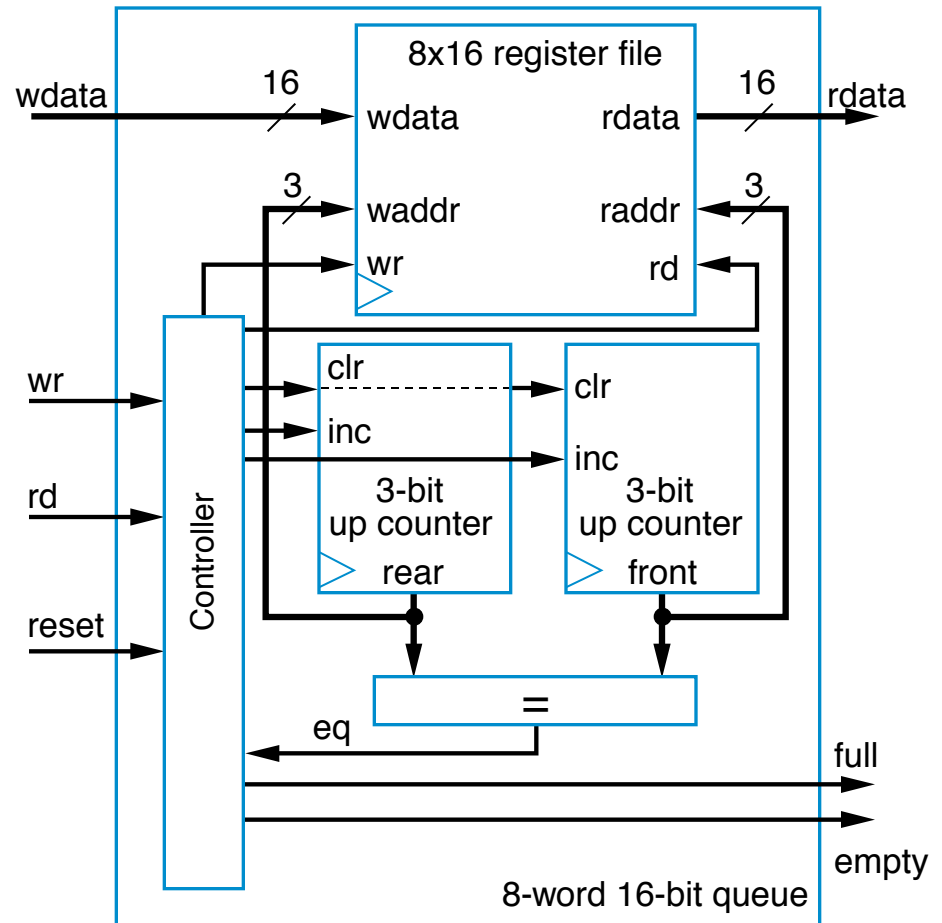
Queues

- Treat memory as a circle
 - If front or rear reaches 7, next (incremented) value should be 0 rather than 8 (for a queue with addresses 0 to 7)
- Two conditions of interest
 - Full queue – no room for more items
 - In 8-entry queue, means 8 items present
 - No further pushes allowed until a pop occurs
 - Causes front=rear
 - Empty queue – no items
 - No pops allowed until a push occurs
 - Causes front=rear
 - Both conditions have front=rear
 - To detect whether front=rear means full or empty, need state machine that detects if previous operation was push or pop, sets full or empty output signal (respectively)



Queue Implementation

- Can use register file for item storage
- Implement *rear* and *front* using up counters
 - rear used as register file's write address, front as read address
- Simple controller would set control lines for pushes and pops, and also detect full and empty situations
 - FSM for controller not shown



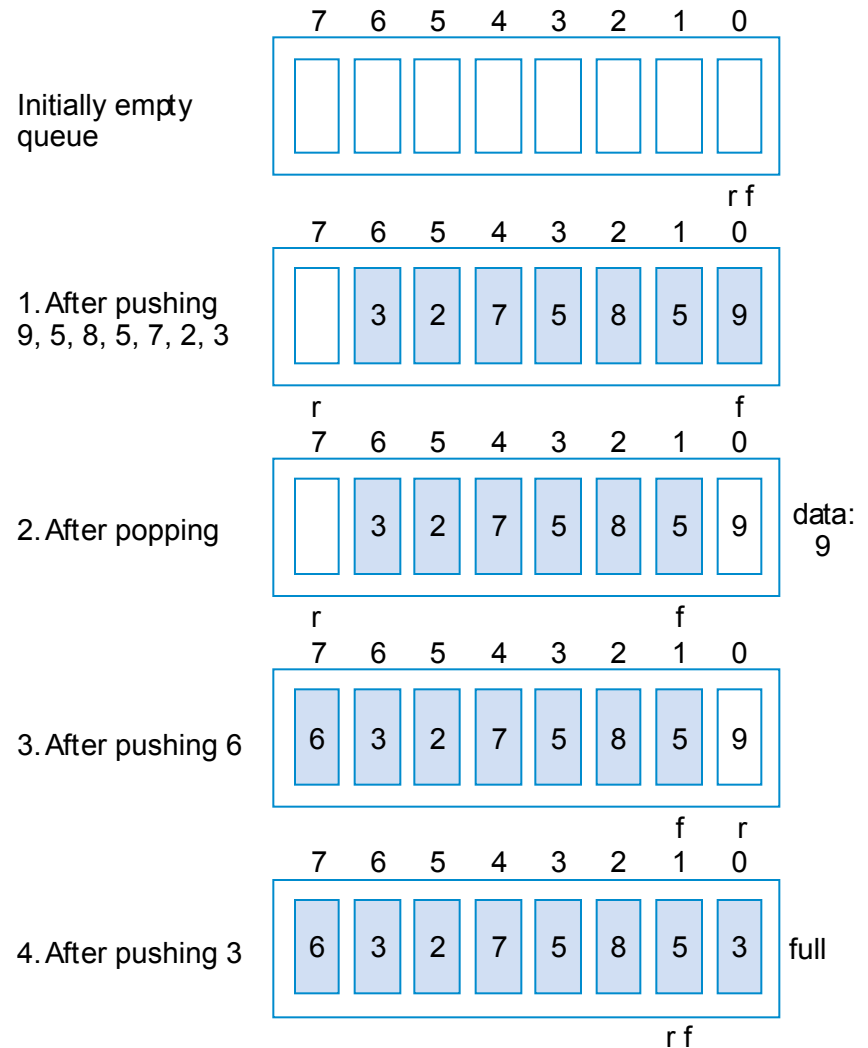
Common Uses of a Queue

- Computer keyboard
 - Pushes pressed keys onto queue, meanwhile pops and sends to computer
- Digital video recorder
 - Pushes captured frames, meanwhile pops frames, compresses them, and stores them
- Computer network routers
 - Pushes incoming packets onto queue, meanwhile pops packets, processes destination information, and forwards each packet out over appropriate port



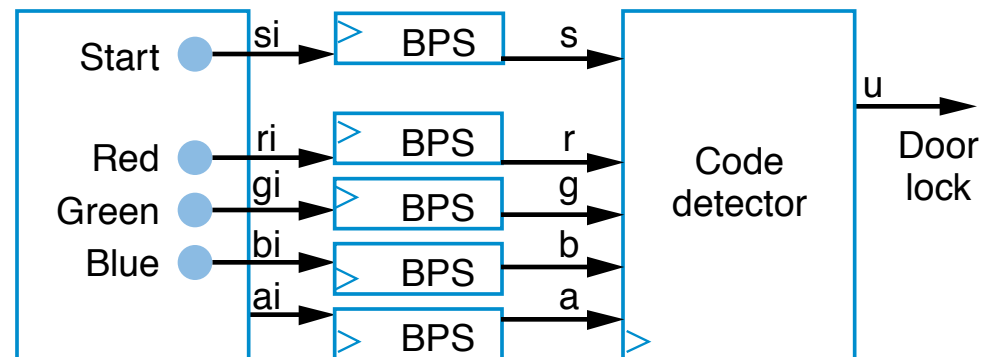
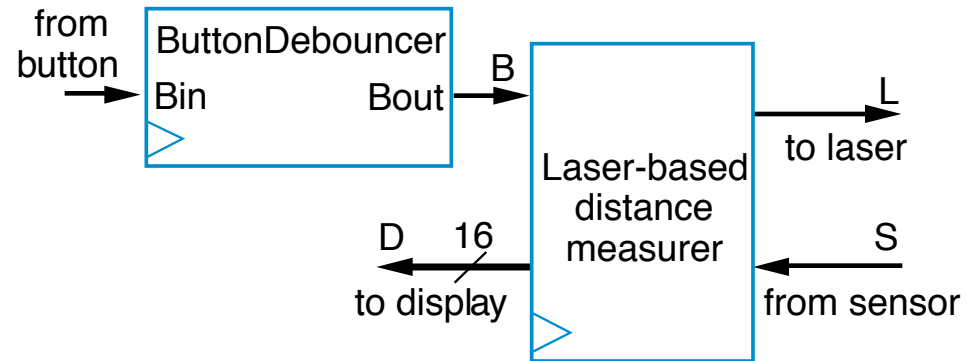
Queue Usage Example

- Example series of pushes and pops
 - Note how rear and front pointers move
 - Note that popping doesn't really remove the data from the queue, but that data is no longer accessible
 - Note how rear (and front) wraps around from address 7 to 0
- Note: pushing a full queue is an error
 - So is popping an empty queue



Multiple Processors

- Using multiple processors can ease design
 - Keeps distinct behaviors separate
 - Ex: Laser-based distance measurer with button debounce
 - Use two processors
 - Ex: Code detector with button press synchronizers (BPS)
 - BPS processor for each input, plus CodeDetector processor



Interfacing Multiple Processors

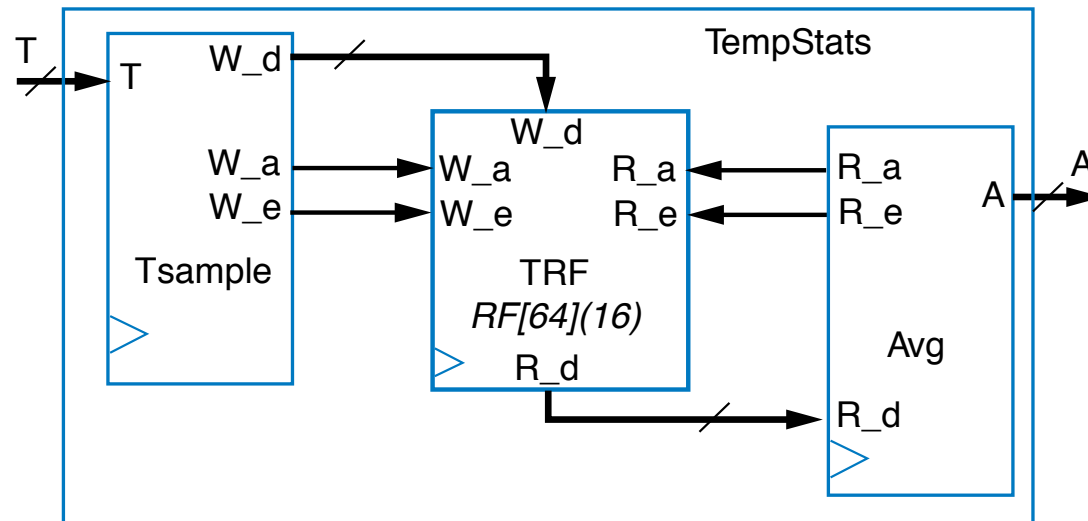
- Use signal, register, or other component outside processors
 - Known as *global*
- Common methods use global...
 - control signal, data signal, register, register file, queue
- Typically all multiple processors and clocked globals use same clock
 - *Synchronized*



Ex: Temperature Statistics with Multiple Processors

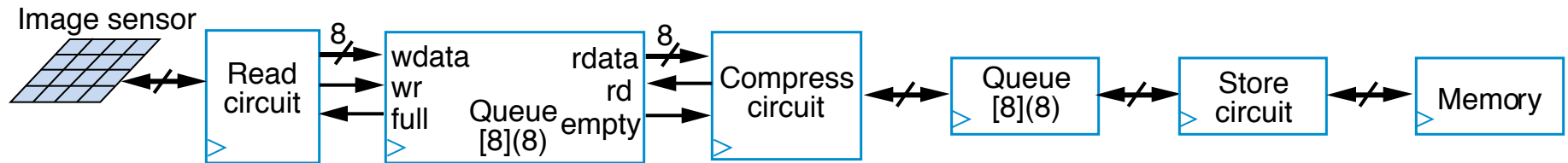
- 16-bit unsigned input T from temperature sensor, 16-bit output A. Sample T every 1 second. Compute output A every minute, should equal average of most recent 64 samples.
- Single HLSM: Complicated
- Instead, two HLSMs (and hence two processors) and shared register file
 - Tsample HLSM: Store T into successive RF address, once per sec.
 - Avg HLSM: Compute and output average of all 64 RF words, once per min.
 - Note that each uses distinct timer

Keeping the sampling and averaging behaviors separate leads to simple design



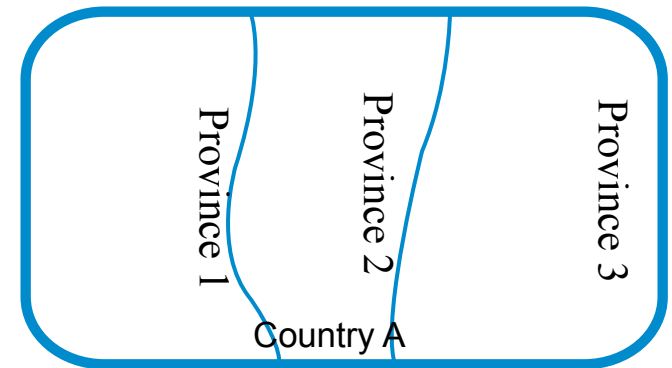
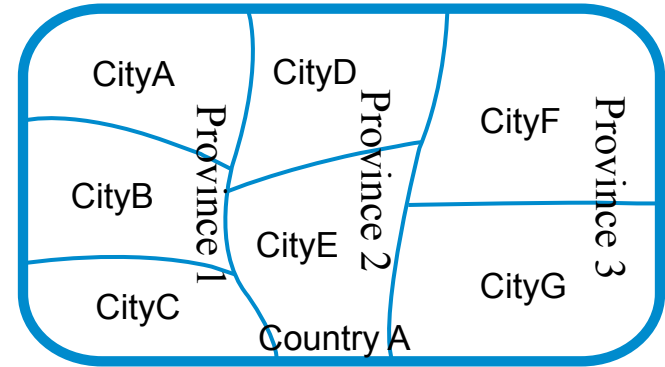
Ex: Digital Camera with Mult. Processors and Queue

- Read and Compress processors (Ch 1)
 - Compress may take longer, depends on picture
 - Use queue, read can push additional pics (up to 8)
 - Likewise, use queue between Compress and Store



Hierarchy – A Key Design Concept

- Hierarchy
 - Organization with few items at the top, with each item decomposed into other items
 - Common example: Country
 - 1 item at top (the country)
 - Country item decomposed into state/province items
 - Each state/province item decomposed into city items
- Hierarchy helps us ***manage complexity***
 - To go from transistors to gates, muxes, decoders, registers, ALUs, controllers, datapaths, memories, queues, etc.
 - Imagine trying to comprehend a controller and datapath at the level of gates

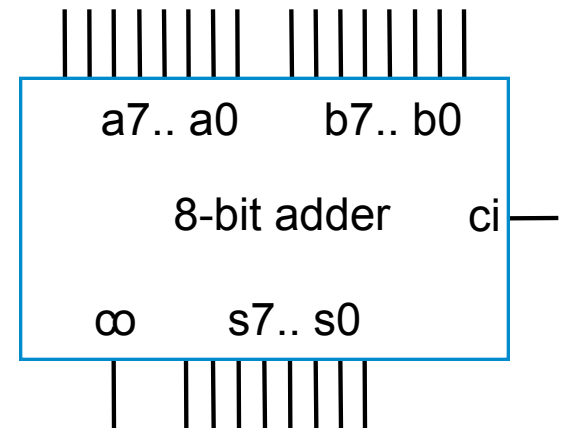


Map showing just top two levels of hierarchy



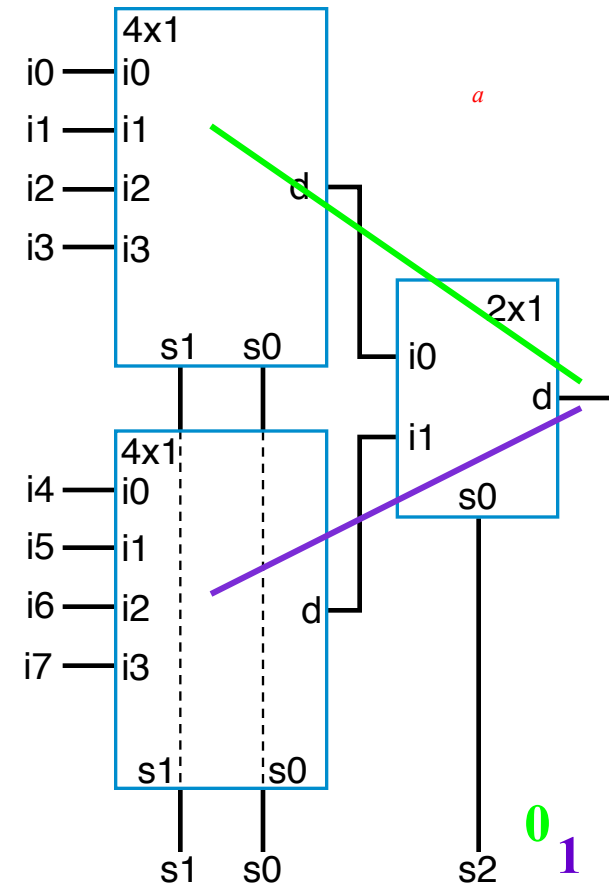
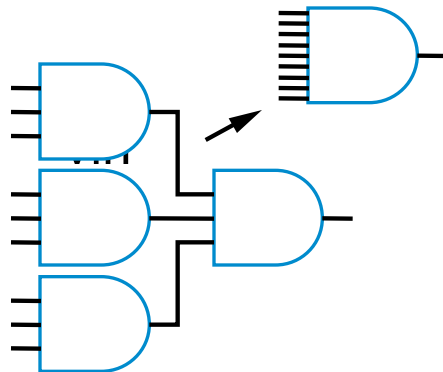
Hierarchy and Abstraction

- Abstraction
 - Hierarchy often involves not just grouping items into a new item, but also associating higher-level behavior with the new item, known as abstraction
 - Ex: 8-bit adder has understandable high-level behavior—adds two 8-bit binary numbers
 - Frees designer from having to remember, or even understand, the lower-level details



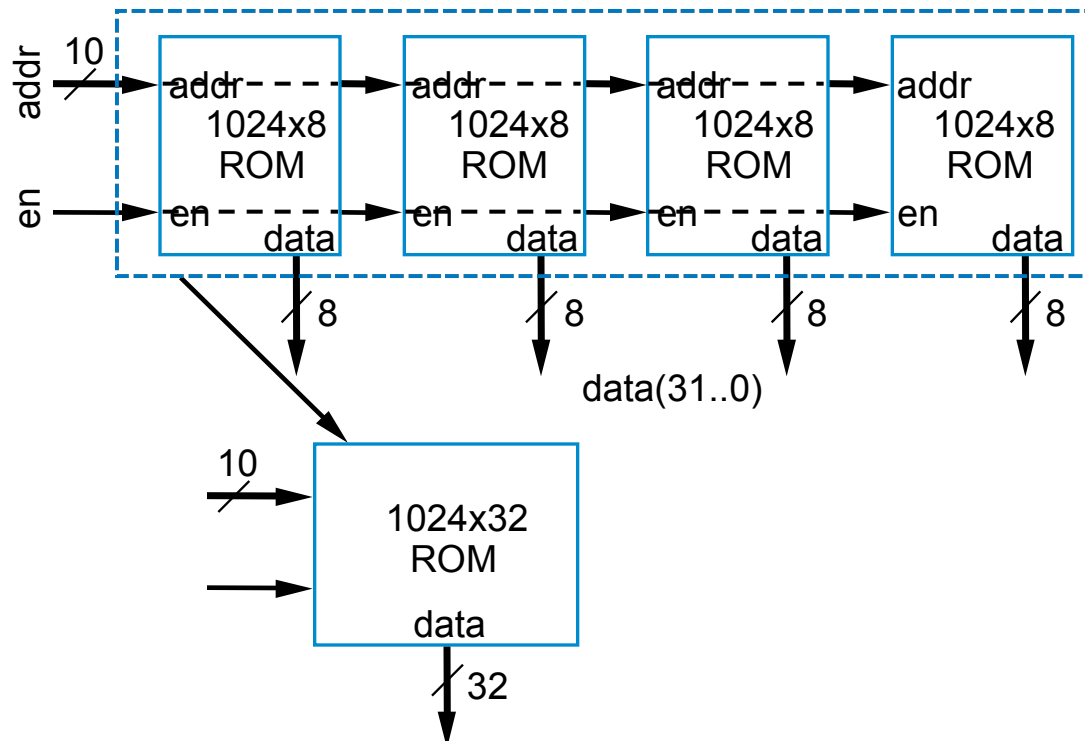
Hierarchy and Composing Larger Components from Smaller Versions

- A common task is to compose smaller components into a larger one
 - Gates: Suppose you have plenty of 3-input AND gates, but need a 9-input AND gate
 - Can simply compose the 9-input gate from several 3-input gates
 - Muxes: Suppose you have 4x1 and 2x1 muxes, but need an 8x1 mux
 - s2 selects either top or bottom 4x1
 - s1s0 select particular 4x1 input
 - Implements 8x1 mux – 8 data inputs, 3 selects, one output



Hierarchy and Composing Larger Components from Smaller Versions

- Composing memory very common
- Making memory words wider
 - Easy – just place memories side-by-side until desired width obtained
 - Share address/control lines, concatenate data lines
 - Example: Compose 1024x8 ROMs into 1024x32 ROM



Chapter Summary

- Modern digital design involves creating processor-level components
- High-level state machines
- RTL design process
 - 1. Capture behavior: Use HLSM
 - 2. Convert to circuit
 - A. Create datapath B. Connect DP to controller C. Derive controller FSM
- More RTL design
 - More components, arrays, timers, control vs. data dominated
- Determining fastest clock frequency
 - By finding critical path
- Behavioral-level design – C to gates
 - By using method to convert C (subset) to high-level state machine
- Memory components (RAM, ROM)
- Queues
- Multiple processors
- Hierarchy: A key concept used throughout Chapters 2-5

